المركز السعودي لسلامة المرضى
SAUDI PATIENT SAFETY CENTER

Patient Safety Standards Book

Version 0.1

"Raising the Bar for Safer Healthcare"

Introduction

The Saudi Patient Safety Center is pleased to present the first draft of the Patient Safety Standards, a significant step forward in the shared commitment to delivering safer healthcare. These standards are an essential milestone aligned with the Centre's core business under article 3—mandate number 2, which is to issue patient safety standards, best practices, and approaches to be employed by healthcare practitioners and organizations.

The Patient Safety Standards took into account the most recent evidence-based practices and the recommended actions and tactics from local and global patient safety institutions, the global patient safety action plan, peer-reviewed patient safety journals, and insights from subject matter experts and healthcare organizations when developing the content of this book.

The skeleton of this standards book is a 10*5*5 framework comprising ten patient safety domains. Five primary criteria characterize each domain, and each criterion is supported by five pieces of evidence contributing to achieving the primary criteria.

The standards book devised a logical and progressive sequence of required evidence to support and advance your patient safety excellence journey. This sequence is designed to help you maintain your current practice by considering each criterion's structure, process, and outcome requirements.

Sharing this draft emphasizes maximizing the extensive collaboration with many stakeholders united by the goal of reducing preventable harm in healthcare settings. Therefore, we invite you to review the document and share your feedback. Together, we can co-create a set of standards that will drive meaningful improvements in patient safety, benefiting patients, families, and healthcare providers alike.

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Domain 1: Quality and Patient Safety Tools

In healthcare, ensuring the safety of patients is a responsibility that permeates every aspect of care delivery [1]. Yet, healthcare organizations face ever-evolving challenges in delivering safe and effective care [2]. Using evidence-based quality and patient safety tools is essential to enhancing patient safety and enabling health professionals to protect those entrusted to their care [3]. The importance of performance improvement tools and models rests on their systematic approach of identifying areas for improvement, prioritizing actions, and monitoring performance progress [4]. Hence, incorporating models such as Lean, Six Sigma, and the Plan-Do-Study-Act (PDSA) cycle can improve patient safety by continually assessing and refining the care processes [5].

Health organizations strive to enhance patient safety through evidence-based tools. Therefore, it becomes imperative to investigate the underlying factors contributing to adverse events [6]. This connection between patient safety tools and root cause analysis forms a crucial synergy, where the insights gleaned from improvement tools guide the in-depth exploration of systemic contributing conditions [7]. Patient safety investigations such as Root Cause Analysis (RCA) is a structured approach to investigate safety events by drilling into the underlying factors that lead to an incident. The goal is to uncover the root causes that, when addressed, can prevent the recurrence of similar events and facilitate lasting improvements [8].

In the academic literature, many other quality and safety tools appear of great impact on patient safety and outcomes. For instance, implementing a standardized communication approach, such as the SBAR (Situation, Background, Assessment, Recommendation), I-PASS (Introduction, Patient, Assessment, Situation, Safety Concerns), or SOAP (Subjective, Objective, Assessment, Plan) promotes clarity, consistency, and accuracy in transmitting patient information [9-11]. This approach can reduce the risk of critical information being lost, miscommunicated, or misinterpreted, leading to improved patient safety and reduced errors during care transitions. Safety huddles are another tool that facilitates regular safety discussions among team members, providing an opportunity to share observations, voice concerns, and propose solutions [12]. Regular huddles help organizations address issues promptly and create a culture that prioritizes vigilance and continuous improvement.

It should be acknowledged, however, that acting to avoid failures prospectively is more valuable compared to reacting to a specific event retrospectively [13]. Therefore, in the pursuit of patient safety excellence, organizations must adopt a proactive improvement approach. This involves identifying, prioritizing, and mitigating potential risks before they escalate into adverse events [14]. Models, such as failure mode and effect analysis (FMEA) and DMADV (Define, Measure, Analyze, Design, Verify) assist healthcare organizations in predicting failures and minimizing the occurrence of preventable safety errors [15].

In conclusion, evidence-based quality and patient safety tools play a critical role in optimizing patient safety. By embracing these tools, healthcare organizations can establish a culture of continuous improvement, open communication, and proactive risk mitigation, ultimately saving more patients and fostering a safer healthcare environment [16].

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Criteria 1.1: the organization uses evidence-based performance improvement models to enhance patient safety.	
Description: *This criterion highlights the organization's commitment to utilizing evidence-based performance improvement models, such as Lean, Six Sigma, PDSA (Plan-Do-Study-Act), or DMAIC (Define-Measure-Analyze-Improve-Control) as a means to enhance patient safety continuously. This subsequently demonstrates a commitment to staying abreast of advancements in the patient safety field and utilizing evidence-based performance improvement approaches as a cornerstone to enhance the quality of service, patient safety, and health outcomes. Healthcare organizations can adhere to these requirements by establishing a written program, policy, or plan outlining their improvement processes, models, prioritizations, and stakeholders involved. Such a document serves as a roadmap to mitigate risks and enhance safety systematically. This plan should be accompanied by records of staff (i.e. clinical and non-clinical) training on performance improvement processes and models. The training on performance improvement is intended for staff as determined by the healthcare organization. Moreover, organizations must provide written evidence of at least two completed patient safety-related improvement projects, with evidence of implementation and involvement of the right interdisciplinary team in these projects. Therefore, patients may be involved in the co-design of these projects. Lastly, there should be evidence of reviewing and monitoring safety practices changes based on the improvement project's findings. This could be through a set of performance indicators, inspection reports, audit rounds, dashboards, committee discussions, leadership discussions, or customer feedback.*	
Evidence:	
1.1.1	*A written performance improvement plan outlines the process, procedure, and models used to enhance the performance.*
1.1.2	*Records of performance improvement training for the staff involved in performance improvement on improvement processes, models, and tools.*
1.1.3	*Documentation of at least two completed and implemented patient safety improvement projects using models such as lean, PDSA, or DMAIC.*
1.1.4	*Evidence of involving a relevant interdisciplinary team in patient safety improvement projects.*
1.1.5	*Evidence of reviewing and monitoring changes in safety practices following performance improvement projects at least quarterly.*

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Criteria 1.2: the organization deploys a systematic approach for investigating patient safety events.
Description: *This criterion highlights the importance of adopting a systematic approach in investigating patient safety events to identify areas for improvement and implement preventive and corrective measures to mitigate the risk and prevent similar incidents in the future. The organization goes beyond merely addressing the immediate factors contributing to an event. It delves deeper into the underlying systemic issues and human factors that may have played a role. A written policy, plan, guideline, or protocol outlining the event investigation process is a foundational guide for healthcare institutions to understand factors contributing to safety incidents. Also, organizations must train the staff involved in the investigation on the processes to equip them with the skills needed to conduct thorough and effective investigations. Such training might be conducted just on time when an event that requires investigation occurs. Safety event reports, completed by interdisciplinary teams using tools like brainstorming, Fishbone diagram, 5-Whys analysis, or tree diagram, serve as tangible evidence of the commitment to pinpointing the root causes. It is required also to submit evidence of developing a corrective and preventive action plan following the investigation. These actions need to be regularly monitored for accomplishment to ensure that the identified improvements are effectively integrated and closed. Such monitoring could be through a set of performance indicators, inspection reports, audit rounds, dashboards, or customer feedback.*

Evidence:	
1.2.1	*A written policy outlines the indications, processes, tools, reporting, responsibilities, and documentation of investigating safety events.*
1.2.2	*Records of training for staff who are involved in the processes of investigating safety events.*
1.2.3	*A report of completed safety event investigation (i.e. within 30 days) by the relevant team using tools such as Fishbone, 5-Whys, or tree diagram.*
1.2.4	*Evidence of developing a corrective action plan following safety event investigation.*
1.2.5	*Evidence of reviewing and monitoring the implementation of the developed action plan based on the investigation findings at least quarterly.*

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Criteria 1.3: the organization establishes a standardized, evidence-based communication approach to reduce handoff errors.	
Description: *This criterion highlights the importance of establishing a standardized, evidence-based handoff approach such as SBAR (Situation, Background, Assessment, Recommendation), I-PASS (Introduction, Patient, Assessment, Situation, Safety Concerns), or SOAP (Subjective, Objective, Assessment, Plan), for exchanging patient care information accurately within healthcare organizations at the provider level (i.e. staff to staff) and patient level (i.e. site to site or unit to unit) to mitigate the risk of handoff errors, promote seamless information exchange, and promote patient safety subsequently. A comprehensive policy delineating the process and tools employed for handoff assists in establishing a standardized language in which healthcare teams communicate, preventing misunderstandings and omissions that can have major consequences. Healthcare organizations need to maintain documented evidence of using such structured handoff tools to promote consistency and thoroughness in conveying critical patient information. Further, An analysis report of reported handoff errors followed by a detailed action plan to reduce them would enable healthcare organizations to learn from mistakes and proactively prevent their recurrence. Lastly, monitoring adherence to handoff policy and documentation helps ensure that established protocols are consistently followed. Such monitoring could be through performance indicators, audit rounds, or chart reviews.*	
Evidence:	
1.3.1	*A policy that outlines handoff processes, procedures, tools, and responsibilities at the provider and patient levels.*
1.3.2	*Evidence of integrating communication and handoff tools in the orientation program of newly hired staff.*
1.3.3	*Documented evidence of using structured, evidence-based tools routinely during handoffs.*
1.3.4	*An analysis report of reported handoff errors and process violations, followed by an action plan to reduce such concerns.*
1.3.5	*Evidence of reviewing and monitoring adherence to handoff policy and documentation at least quarterly.*

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Criteria 1.4: the organization utilizes a structured safety huddle tool to facilitate raising and reporting patient safety concerns.	
Description: *This criterion emphasizes adopting a structured safety huddle tool at the organizational and departmental levels to facilitate identifying, raising, and reporting patient safety concerns. The tool serves as a purposeful communication mechanism to enable health professionals to share their observations, potential hazards, and other safety-related issues in a focused, coordinated, and transparent manner to lead a safer healthcare delivery. A policy outlining the purpose, procedure, frequency, and participants of safety huddles is a cornerstone to ensure clarity and consistency of safety huddle implementation. Also, healthcare organizations should invest in training encouraging active participation from all relevant interdisciplinary teams in safety huddles. Such training is essential to ensure all participants have the knowledge and skills to contribute effectively. Further, keeping documented evidence of training records and evidence of using safety huddle tools is important for tracking and analysis purposes. An analysis report of raised safety concerns through safety huddles, followed by action plans, signifies a commitment to addressing and preventing identified existing vulnerabilities promptly. Lastly, monitoring adherence to safety huddle practices using performance indicators, audit rounds, or staff feedback reports is crucial in ensuring the sustainability and effectiveness of this safety initiative.*	
Evidence:	
1.4.1	*A written policy outlines the safety huddle's purpose, process, procedure, responsibilities, and frequency.*
1.4.2	*Records of safety huddle training for staff involved in the safety huddle process.*
1.4.3	*Documented evidence (i.e. report, log, meeting minutes) of using safety huddles tool or procedure.*
1.4.4	*An analysis report of raised safety concerns through safety huddles, followed by documented action plans.*
1.4.5	*Evidence of reviewing and monitoring adherence to safety huddles policy and documentation at least quarterly.*

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Criteria 1.5: the organization employs a systematic, proactive approach to identify, prioritize, and mitigate potential risks.	
Description: *This criterion emphasizes the importance of establishing a systematic risk management framework to identify, evaluate, prioritize, and effectively reduce or mitigate potential risks in healthcare settings. This framework assists in identifying potential risks that may impact patient safety or compromise operational efficiency or the overall quality of care. After risk identification, the healthcare organization adopts an evidence-based grading/scoring system that is based on the likelihood and consequences of these risks to help in prioritizing them, such as risk priority number (RPN), hazard and operability study (HAZOP) or severity assessment code (SAC). Thereafter, a methodical prospective approach such as Failure Mode and Effects Analysis (FMEA) or DMADV (Define, Measure, Analyze, Design, Verify) is employed to develop decisive actions addressing these risks systematically. To streamline risk management and proactive risk reduction processes, healthcare organizations need to train the involved personnel to detect risks and use such proactive approaches. Such training is anticipated to equip personnel involved in risk reduction projects with the necessary knowledge and skills. Lastly, the ongoing review and monitoring of implemented risk reduction strategies ensure that the organization's risk management efforts remain relevant and effective. The monitoring could be through a set of performance indicators, inspection reports, audit rounds, dashboards, or customer feedback.*	
Evidence:	
1.5.1	*A written risk management plan or guideline outlines the processes of risk identification, evaluation, prioritization, and treatment.*
1.5.2	*Records of risk management training for relevant staff involved in risk evaluation and reduction processes and projects.*
1.5.3	*A report outlines organizational risks that are identified and prioritized based on their severity and likelihood using a risk scoring tool.*
1.5.4	*Evidence of employing a proactive risk reduction approach, such as FMEA or DMADV, by a relevant interdisciplinary team with implemented actions.*
1.5.5	*Evidence of reviewing and monitoring risk reduction strategies implemented based on the prioritized risks at least quarterly.*

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Domain 2: Leadership Commitment to Patient Safety

In rapidly evolving healthcare, the great concern for health organizations is the safety and well-being of their patients [17]. Leaders play a pivotal role in shaping a culture of dedication to patient safety that stimulates the successful implementation of patient safety initiatives. Leaders dedicated to patient safety demonstrate a steadfast commitment to safety as a top priority, openly communicating safety goals and actively engaging with staff and patients in safety initiatives [18]. Through their dedication and guidance, leaders establish a solid foundation for providing safe care, create a resilient culture of safety that safeguards patients, inspires trust, fosters a sense of purpose among staff, and drives continuous improvement in healthcare delivery, which ultimately leads to better outcomes and satisfaction [19].

Effective leadership commitment to safety starts with ensuring that patient safety is a central priority. In this, leaders set the tone by officially designating patient safety as a strategic priority, reflected in mission statements, strategic plans, and policies emphasizing patient safety [20]. Further, dedication to patient safety translated this strategic priority into fundamental actions such as participating in a structured safety leadership WalkRounds™ program [21]. In this program, leaders establish open and transparent communication with frontline staff, fostering an environment where safety concerns are identified, discussed, and acted upon collaboratively [21].

In addition to patient safety concerns revealed during WalkRounds™ sessions, excellent practices are noticed and should not be overlooked without being rewarded. Therefore, healthcare organizations need to create a rewarding system to recognize staff members who actively contribute to patient safety through reporting or participation in patient safety initiatives [22]. At the individual and departmental levels, the reward system encourages and motivates staff to engage in patient safety efforts [23]. The reward might be achieved at the organizational level by pursuing and obtaining a recognized national or international patient safety award and certificate [24]. These criteria demonstrate the organization's dedication to meeting and exceeding patient safety standards, inspiring confidence among patients, families, and the broader healthcare community.

It should be acknowledged, however, that leaders hold various levels of patient safety competency [25]. Hence, healthcare organizations need to clearly define the knowledge and skills necessary for leaders to communicate, collaborate, and lead by example in promoting patient safety in every aspect of care delivery [26].

In conclusion, leadership commitment to patient safety goes beyond words; it is exemplified by taking actions and initiatives that prioritize the safety of patients and the excellence of care. By designating patient safety as a central strategic priority, implementing structured WalkRounds™ programs, creating a rewarding system, addressing leadership competencies, and pursuing patient safety recognition, leadership promotes a culture where patient safety is not just a goal but a shared commitment that resonates throughout the organization.

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Criteria 2.1: the organization demonstrates its commitment to safety by designating patient safety as a central strategic priority.	
Description: *This criterion highlights the mechanism in which the organization exemplifies dedication to patient safety by designating patient safety as a central strategic priority to enhance safety culture at all levels, from the organizational board to the frontline staff members. In this, healthcare organizations strategically prioritize patient safety in their projects, plans, meetings, and daily business operations. Developing a patient safety plan aligned with the organizational strategic priority is essential to further reinforce patient safety. This plan outlines the organizational efforts to improve patient safety and reduce errors. The plan must be reviewed and approved by the organizational director/chief executive officer (CEO) to underscore the significance of patient safety from the highest levels of leadership. Furthermore, regular discussions of safety reports and proactive solutions during leadership and board meetings are vital. These discussions and participation of leaders in the patient safety journey promote a transparent and responsive approach to addressing safety concerns, fostering a learning environment, and assisting leaders in supporting patient safety projects and initiatives. Lastly, the routine review of patient safety indicators by relevant leaders at least quarterly provides continuous monitoring and insight into the organization's safety performance, enabling timely adjustments to strategies and interventions.*	
Evidence:	
2.1.1	*A written strategic plan that integrates patient safety as an organizational strategic priority.*
2.1.2	*A written organizational patient safety plan that is reviewed and approved by the organizational director/chief executive officer (CEO).*
2.1.3	*Evidence of discussing or suggesting solutions based on safety reports or major patient safety events in leadership and board meetings.*
2.1.4	*Evidence of leadership participation in at least two patient safety projects, events, or initiatives.*
2.1.5	*Evidence of reviewing and monitoring the results of patient safety indicators by relevant leaders at least quarterly.*

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Criteria 2.2: the organization develops and implements a structured patient safety leadership WalkRounds™ program to promote safety.	
Description: *This criterion highlights the importance of strategically employing patient safety leadership WalkRounds™ to strategically advance patient safety in healthcare organizations. This initiative entails the active participation of senior leaders and clinical department heads in regular rounds, engaging with frontline staff to assess safety culture, identify potential safety concerns, and promote free reporting of safety concerns. For consistency, it is essential to establish a policy that outlines the objectives, scope, frequency, and assigned responsibilities of patient safety leadership WalkRounds™. This policy serves as the foundational document that clarifies the entire process. Moreover, healthcare organizations must train their executives and leaders on leadership WalkRounds™ to equip them with the necessary knowledge and skills to conduct these rounds without drifting from the intended purpose. To ensure the regularity of these rounds, organizations must also prepare a schedule for monthly patient safety leadership WalkRounds™. Such a schedule may include details on dates, participants, and areas planned to be visited. Moreover, healthcare organizations must maintain documented evidence that details the discussions and action plans developed to address the safety issues raised during these rounds. Subsequently, healthcare organizations must review and monitor action plans to ensure they are effectively implemented. This monitoring could be through a set of performance indicators, inspection reports, audit rounds, dashboards, or customer feedback reports.*	
Evidence:	
2.2.1	*A policy outlines the process, procedure, frequency, and responsibilities of the patient safety leadership WalkRounds™.*
2.2.2	*Records of training organization's executives and leaders on patient safety leadership WalkRounds™.*
2.2.3	*Records of conducting patient safety leadership WalkRounds™ at least monthly, including participants, visited areas, and the voiced safety issues.*
2.2.4	*Documented evidence of developing and implementing action plans following leadership WalkRounds™ to address voiced safety issues.*
2.2.5	*Evidence of reviewing and monitoring the implementation of action plans that were developed following patient safety leadership WalkRounds™.*

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Criteria 2.3: the organization's leaders create a rewarding system to recognize staff members who actively contribute to patient safety.	
Description: *This criterion highlights the dedication of leaders toward patient safety by instituting an internal reward policy designed to commend staff members who are actively engaged in advancing the quality of care. The organization must integrate patient safety contributions as a recognized effort that requires acknowledgment and appreciation. The policy aims to enhance the safety culture and motivate healthcare professionals by identifying individuals who consistently and positively contribute to promoting patient safety. For fairness and understanding purposes, the policy shall include the selection and evaluation criteria for recognizing patient safety contributions. Also, this policy shall be communicated to all staff members to encourage widespread participation. Moreover, healthcare organizations should show instances where leadership has formally recognized staff members who have acquired advanced patient safety certifications, actively reported safety concerns, or consistently engaged in safety initiatives. This recognition acknowledges individual commitment to enhancing patient safety and sets a precedent for others. Furthermore, healthcare organizations should review and monitor compliance with and the impact of implementing the rewarding policy. This monitoring could be through assessment reports, culture surveys, satisfaction surveys, or staff feedback reports. Compliance with this criterion is an administrative necessity that promotes a patient-centric ethos within healthcare organizations.*	
Evidence:	
2.3.1	*A written rewards policy that integrates patient safety contributions as an effort requires organizational acknowledgment and appreciation.*
2.3.2	*Evidence of communicating the selection and evaluation criteria of recognizing patient safety contributions to all organizational staff.*
2.3.3	*Evidence on instances demonstrates leadership recognition for staff who acquired advanced patient safety certification.*
2.3.4	*Evidence on instances demonstrates leadership recognition for staff active in reporting safety concerns or consistently involved in safety initiatives.*
2.3.5	*Evidence of reviewing and monitoring compliance with and impact of implementing the rewarding policy at least quarterly.*

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Criteria 2.4: the organization defines the competencies required for its leaders to champion patient safety initiatives effectively.

Description:

This criterion underscores the necessity of reviewing and evaluating the skills and competencies of its leaders to ensure their ability to drive patient safety initiatives effectively. In this, healthcare organizations need to establish a framework that outlines the competencies required for leaders to champion patient safety initiatives effectively. This competency framework serves as a strategic roadmap, defining the skills, knowledge, and responsibilities leaders need to lead by example, ensuring patient safety remains a core priority. Hence, integrating these competencies into the job descriptions of organizational leaders would foster patient safety as a fundamental part of their duties. Similarly, organizations should utilize patient safety-related competencies in organizational leaders' recruitment and evaluation process. For instance, assessing leader candidates based on patient safety competencies during interviews ensures that the leader possesses the qualifications and mindset to drive patient safety initiatives forward. Lastly, healthcare organizations must regularly review and monitor the perceptions of both staff and patients regarding the leadership's effectiveness in championing patient safety. Such monitoring could be obtained through a designated survey or part of other surveys such as staff satisfaction, patient satisfaction, and safety culture surveys. The feedback allows for the continuous improvement of leadership practices and highlights the importance of a culture of accountability and transparency.

Evidence:

2.4.1	*A written competency framework outlines the required competencies for leaders to champion patient safety initiatives.*
2.4.2	*Evidence of integrating patient safety roles, responsibilities, knowledge, and skills in the job descriptions of organizational leaders.*
2.4.3	*Documented evidence that leaders use patient safety-related competencies in addressing and responding to patient safety events.*
2.4.4	*Evidence of appraising leaders' performance annually based on patient safety metrics and indicators.*
2.4.5	*Evidence of reviewing and monitoring the perception of staff and patients regarding the leadership's effectiveness in championing patient safety.*

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Criteria 2.5: the organization demonstrates its dedication to patient safety by pursuing and achieving external patient safety recognition.	
Description: *This criterion emphasizes the value of dedication to patient safety through pursuing and accomplishing national or international patient safety awards or certifications, such as the SPSC National Patient Safety Award (NPSA) and Patient Safety-Friendly Hospital Initiative (PSFHI). Participation or applying for patient safety certificates or awards is an initial level demonstrating a commitment to patient safety. However, obtaining such certificates or awards is a step ahead of the level of dedication and unique implementation of safety rules. Healthcare organizations need to keep documented evidence of applying for, participating in, and obtaining such certificates or awards. Equally important, organizations need to conduct awareness campaigns for staff regarding the specific requirements of these safety awards or certificates, ensuring that every team member understands their role in meeting the criteria and contributing to patient safety. Furthermore, healthcare organizations should transparently communicate the pursuit and achievement of patient safety recognition internally and externally, celebrating these accomplishments with staff and sharing them with patients, families, and the broader community. Last but not least, maintaining the performance gains preceding or following the attainment of safety recognition is important. Hence, regular review and monitoring of metrics that demonstrate ongoing improvements in patient safety outcomes provide a quantifiable measure of the positive impact of these certificates or awards.*	
Evidence:	
2.5.1	*Documented evidence of applying for or participating in national or international patient safety awards or certificates.*
2.5.2	*Records of providing training or awareness sessions to the staff regarding safety awards or certificate requirements.*
2.5.3	*Evidence of obtaining a national or international patient safety award or certificate.*
2.5.4	*Evidence of communicating the pursuit and achievement of patient safety recognition internally and externally.*
2.5.5	*Evidence of reviewing and monitoring improvements in patient safety outcomes preceding or following the safety recognition.*

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Domain 3: Patient Safety Culture

Patient safety culture has emerged as a guiding path toward safer and more compassionate care in healthcare. Patient safety culture represents the collective values, beliefs, and behaviors that shape the safety-conscious mindset of an organization [27]. A positive patient safety culture is vital in saving lives, promoting outcomes, and enhancing how healthcare is practiced and perceived [28]. Therefore, assessing safety culture helps organizations identify areas for improvement and enable the evolution of meeting patient safety challenges.

In patient safety, the influence of leadership cannot be overstated. From boardrooms to bedside, leaders play a role in setting the tone, aligning priorities, and empowering frontline caregivers, subsequently nurturing a safety culture [29]. In addition, other factors assist in shaping the safety culture, such as organizational structure, safety policies, communication, and patient engagement [30]. It should be acknowledged, however, that patient safety culture is confronted by diversified barriers and challenges, such as time pressures, workload, and resistance to change [31]. Hence, healthcare organizations need to adopt measures to enhance the culture of patient safety, such as developing a patient safety plan, forming an interdisciplinary committee overseeing quality and patient safety, and conducting periodic patient safety awareness campaigns.

Using validated tools for measuring staff perceptions of patient safety culture is important [32]. It ensures that the healthcare organization understands its safety culture, areas of strength, and areas that require improvement [33]. Furthermore, an incident reporting system encourages open communication about safety concerns and allows organizations to identify potential risks [34]. However, voluntary incident report systems may not detect all errors. Thus, it is recommended to use a combination of various tools to detect medical errors and learn from them [35]. For instance, effectively employing trigger tools to capture missed or unreported safety events is critical [36]. These tools serve as an additional layer of defense in the pursuit of patient safety, enabling organizations to detect errors that might otherwise remain hidden.

Transparency and learning from errors create an environment where individuals feel comfortable reporting safety events without fear of retribution [37]. In this, just culture guides the way toward fairness, accountability, and resilience, eventually empowering individuals to freely report errors, knowing that they will be treated fairly and that the focus is on learning and system improvement rather than blame [38].

In the modern era of interconnected healthcare systems, collective efforts to pursue and advance patient safety culture by sharing successful models and best practices in patient safety remain a common thread that unites the global healthcare community. In Saudi Arabia, the health system has recognized the importance of integrating patient safety principles into healthcare policies and practices. Subsequently, significant strides have been taken to foster a positive safety culture across healthcare institutions.

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Criteria 3.1: the organization adopts measures to enhance patient safety culture.	
Description: *This criterion demonstrates the organization's commitment to fostering a patient safety culture through comprehensive measures. These measures might range from launching a patient safety awareness campaign to adopting artificial intelligence technologies that complement safety practices. Therefore, documenting planned patient safety activities and initiatives in an annually revised patient safety plan encourages alignment with organizational patient safety objectives. This plan must be tailored to the emerging patient safety issues, errors, and reported safety concerns. Achieving patient safety objectives without involving relevant stakeholders appears impossible. Hence, launching patient safety campaigns annually plays a vital role in translating these objectives into actionable awareness. These campaigns raise staff awareness about patient safety goals, measures, and best practices. Furthermore, the evidence of regular discussions on patient safety policies, initiatives, and indicators within an interdisciplinary patient safety committee serves as a dynamic platform for collaboration and knowledge sharing. Crucially, allocating resources such as staffing, training, and technology to support patient safety culture and initiatives demonstrates a tangible commitment to enhancing patient safety. Finally, reviewing and monitoring adherence to patient safety practices brings a sense of accountability and continuous improvement. This monitoring could be through performance indicators, audit rounds, tracers, or chart reviews.*	
Evidence:	
3.1.1	*A written annual patient safety plan that outlines the planned activities and initiatives to prioritize and enhance patient safety.*
3.1.2	*Records of launching patient safety campaigns to raise staff awareness of patient safety goals, measures, and practices at least annually.*
3.1.3	*Evidence of discussing patient safety policies, initiatives, and indicators in an interdisciplinary patient safety committee at least quarterly.*
3.1.4	*Evidence of allocating resources to support organizational patient safety culture and initiatives.*
3.1.5	*Evidence of reviewing and monitoring adherence to patient safety policies, measures, and goals at least quarterly.*

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Criteria 3.2: the organization adopts a validated tool to measure staff perceptions of patient safety culture and take actions accordingly.	
Description: *This criterion highlights how a healthcare organization is committed to providing safe care by using an evidence-based validated tool annually to assess the perceived safety culture among clinical and non-clinical staff. The evidence-based assessment allows the organization to identify strengths, areas for improvement, and trends over time, laying the foundation for focused and effective safety culture initiatives. However, healthcare organizations need to administer the assessment tool to a representative number of staff members to ensure an accurate understanding of the safety culture within the organization. After administering the survey, analyzing survey data at organizational, departmental, and professional levels and sharing the survey results with participating staff through emails, bulletin boards, posters, or meeting discussions demonstrate transparency and accountability while providing valuable insights into areas that require attention. Thereafter, healthcare organizations need to develop a time-framed action plan that addresses at least the lowest three patient safety culture domains or more to ensure that identified weaknesses are acknowledged and addressed. Lastly, reviewing and monitoring progress in implementing the patient safety culture action plan at least quarterly reflects a commitment to continuous improvement. These initiatives serve as actions to foster an environment where the culture of safety is a priority.*	
Evidence:	
3.2.1	*Evidence of using an evidence-based validated tool to assess safety culture aligned with the written patient safety plan, at least annually.*
3.2.2	*Evidence of administering and collecting the patient safety culture survey to a representative sample from all staff (clinical and non-clinical).*
3.2.3	*Evidence of analyzing safety culture data and sharing the analysis results with participating staff (e.g. email, poster, meeting minutes).*
3.2.4	*Evidence of developing a time-framed action plan addressing at least one of the three lowest-scored patient safety culture domains.*
3.2.5	*At least quarterly, evidence of reviewing and monitoring the progress in implementing the patient safety culture action plan.*

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Criteria 3.3: the organization utilizes an incident reporting system to promote reporting and learning safety culture.	
Description: *This criterion highlights establishing an incident reporting system to foster a reporting and learning safety culture within the healthcare organization. Such systems (i.e. electronic or paper-based) encourage staff to report incidents, errors, and near-misses openly and transparently. These reports are instrumental in guiding targeted interventions to prevent future occurrences and learning continuously. To define this process, healthcare organizations need to establish a written incident reporting policy to standardize the reporting approach. The policy should outline reporting indications, methods, and timeframes to ensure that staff members understand the procedures and expectations for incident reporting. Furthermore, training all organizational staff on the importance of incident reporting and the use of the reporting system is vital to ensure that all staff members are aware of their role in identifying and reporting safety incidents and to encourage active reporting by all disciplines. Following reporting, healthcare organizations need to develop action plans in response to reported incidents based on a severity matrix such as the severity assessment code (SAC), followed by providing feedback to reporters to keep them informed of actions taken to address their reported event. Lastly, reviewing and monitoring the frequencies, categories, and trends of reported safety incidents and implementing action plans is crucial for data-driven improvements.*	
Evidence:	
3.3.1	*A written incident reporting policy that outlines the reporting indications, methods, responsibilities, and time frame, as well as the analysis and feedback procedures.*
3.3.2	*Records of staff training on the importance of incident reporting and the use of the reporting system.*
3.3.3	*Records of active reporting by all disciplines submitted via a standardized incident reporting system (i.e. electronic or paper-based)*
3.3.4	*Evidence of developing action plans in response to reported incidents based on the severity matrix, followed by sending feedback to reporters.*
3.3.5	*Evidence of reviewing and monitoring frequencies, categories, and trends of reported safety incidents and implementing action plans quarterly.*

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Criteria 3.4: the organization employs trigger tools to effectively and accurately capture missed or unreported safety events and errors.

Description:

This criterion underscores the importance of employing trigger tools in the error surveillance approach. The Institute for Healthcare Improvement (IHI) global trigger tools are one of the most commonly used tools. The idea is to use predefined "triggers" that suggest the possibility of an adverse event. These triggers prompt conducting a thorough medical records review to determine if an adverse event has occurred but is missed or unreported through traditional reporting systems. Healthcare organizations need to establish a written policy, protocol, or guideline that defines the triggers and the process for reviewing identified cases to ensure that staff members clearly understand the working mechanism of trigger tools. Hence, the organization must specify the selection and definition of each trigger tool to ensure consistent analysis. Further, staff members involved in the trigger process, as determined by the healthcare organization, need to be trained on the effective use, review, and validation of trigger findings to equip them with the knowledge and skills needed to optimally leverage trigger tools. Proper training ensures that potential safety incidents are not overlooked and that identified cases are thoroughly reviewed. Lastly, reviewing and monitoring trigger tool reviews to identify patterns, trends, and areas of concern provides valuable insights into recurring safety issues. It allows the organization to target its resources and efforts where they are most needed, ultimately enhancing patient safety.

Evidence:

3.4.1	*A written protocol or guideline defines the process, procedure, and responsibilities in which triggers are used and how identified cases are reviewed.*
3.4.2	*A training record on how to use, review, and validate trigger findings effectively for staff involved in the trigger process.*
3.4.3	*A record indicates the organization's consistent and systematic use of clinical trigger tools.*
3.4.4	*Evidence demonstrates instances where cases are flagged by trigger tools and reviewed to determine if an error occurred.*
3.4.5	*Evidence of quarterly reviewing and monitoring trigger tool reviews, identifying patterns, trends, and areas of concern.*

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Criteria 3.5: the organization employs a just culture approach to foster a safe, fair, and more accountable healthcare environment.	
Description: *This criterion underscores that the just culture approach is based on the belief that everyone makes mistakes but that not all are equal. In this, individuals are held accountable for their reckless or intentional behavior while providing support and learning opportunities for those who make inevitable human errors. To govern this process, healthcare organizations need to establish a just culture policy and algorithm that is aligned with the organizational priorities and the national just culture framework. This policy and algorithm provide a standardized, evidence-based approach to analyzing and categorizing individual behaviors in the context of errors. In just culture, errors are categorized into human errors that require supporting the staff (i.e. second victim), at-risk behaviors that require coaching, and reckless behaviors that require remedial and possible disciplinary actions. Further, training leaders, departmental heads, and managers on just culture is essential to equip them with the knowledge and skills needed to implement the principles of just culture, reduce subjectivity, and ensure that actions taken are proportionate to the nature of the behavior. Lastly, reviewing and monitoring the impact of implementing the just culture policy on patient safety culture and incident reporting is crucial for assessing the policy's effectiveness. The review could be based on indicators, surveys, or audits to measure the overall improvement of safety outcomes.*	
Evidence:	
3.5.1	*A written just culture policy defines and regulates the analysis of individual behaviors related to errors, which aligns with the national just culture policy.*
3.5.2	*Records of training for leaders, departmental heads, and managers on just culture principles.*
3.5.3	*Evidence of adopting and utilizing an evidence-based just culture algorithm to guide the analysis of behavioral choices.*
3.5.4	*Evidence of using staff support programs such as the "second victim" principle to support staff experienced emotional trauma following human errors as per just culture policy.*
3.5.5	*Evidence of reviewing and monitoring the impact of implementing just culture principles on patient safety culture and incident reporting.*

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Domain 4: Workforce Safety

Workforce safety is a critical concern in healthcare settings, where health professionals tirelessly provide essential care to patients while navigating diverse challenges [39]. Ensuring the safety and well-being of healthcare staff is not only imperative for ethical reasons but also crucial for maintaining high-quality patient care and promoting organizational effectiveness, particularly with the fast-paced and emotionally demanding nature of the healthcare industry [40]. One of the crucial aspects of workforce safety is maintaining a safe staffing level to deliver high-quality patient care. Insufficient staffing can increase workloads, burnout, and compromised patient safety [41]. Hence, healthcare organizations optimize staff-to-patient ratios proactively to reduce the risk of medical errors and adverse events. Therefore, safe staffing levels enable healthcare professionals to provide timely and effective care, fostering a positive work environment that prioritizes patient and staff safety [42].

As an integral part of workforce safety, introducing programs such as the workplace violence prevention program, occupational health and safety program, and work-life balance program appears to be of great value and is linked to patient and staff outcomes [43]. Workplace violence poses a significant threat to the safety and well-being of healthcare staff [44]. Implementing workplace violence prevention programs is necessary to protect employees from potential harm. These programs include strategies such as training staff to recognize warning signs, de-escalation techniques, and appropriate responses to violent incidents [45]. Organizations could create a secure environment by actively addressing workplace violence, boosting staff confidence, and ensuring they can focus on providing optimal patient care.

In the same vein, the occupational health and safety program plays a vital role in safeguarding healthcare staff from work-related risks and promoting their well-being [46]. The program involves comprehensive risk assessments to identify potential hazards and the implementation of measures to prevent injuries and illnesses. Also, introducing work-life balance programs supports staff in achieving harmony between their personal and professional lives [47]. By promoting a positive work-life balance, organizations can retain skilled staff, improve productivity, and foster a culture of resilience [47]. Furthermore, adopting the Practice Partnership Model of Care (PPM) contributes to workforce safety through collaboration among healthcare professionals, promoting effective communication, and coordinated efforts to deliver comprehensive patient care close to patients' bedside [48].

In conclusion, to ensure the success of workforce safety mechanisms, healthcare organizations must integrate all these initiatives and programs cohesively. A holistic approach to workforce safety enhances overall staff satisfaction, mitigates staff turnover, and creates an environment where patient care thrives. Leadership support and ongoing evaluations are essential to continuously improve and sustain these mechanisms, creating a safe, supportive, and high-performing healthcare environment for staff and patients.

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Criteria 4.1: the organization maintains a safe staffing level to ensure the delivery of safe and efficient patient care.	
Description: *This criterion places great importance on maintaining a safe staffing level as a fundamental pillar to ensuring the delivery of safe care. By optimizing staff-to-patient ratios, the organization seeks to uphold a healthy environment where healthcare professionals can dedicate focused attention, expertise, and time to each patient's needs. Adequate staffing levels enable timely interventions, close monitoring, and reducing the risk of errors while maximizing health outcomes. Hence, healthcare organizations should develop a written staffing plan tailored to ensure adequate staff-to-patient ratios, considering patients' acuity and care complexity. This plan should meet regulatory requirements and consider the unique needs of the patient population served. Also, the plan should be complemented by a guideline regulating staff scheduling and workload, including scheduling, overtime, work hours, and rest breaks to avoid staff overburden and fatigue-related risks. In staff shortage cases, healthcare organizations should adopt various temporary measures to overcome this shortage, such as employing overtime, cross-training, or adopting flexible staffing models. Further, healthcare organizations must verify that agency staff, if available, are appropriately trained and licensed as per applicable national regulations. Lastly, reviewing and monitoring staffing levels at least quarterly while assuming corrective actions to address shortages allows organizations to respond to staffing challenges and maintain safe care.*	
Evidence:	
4.1.1	*A written staffing plan to ensure adequate staff-to-patient ratios based on patients' acuity and care complexity as per the relevant authorities.*
4.1.2	*A written policy, plan, or guideline that regulates staff scheduling and workload, including work hours and rest breaks.*
4.1.3	*Evidence of implementing measures to address the identified staffing shortages, such as overtime, cross-training, or flexible staffing models.*
4.1.4	*Evidence to ensure that agency (i.e. outsourced) staff are licensed, trained, and competent as per their job description, if applicable.*
4.1.5	*Evidence of reviewing and monitoring staffing levels at least quarterly and taking corrective actions to address staff shortage as necessary.*

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Criteria 4.2: the organization develops and implements workplace violence prevention programs to protect healthcare staff.	
Description: *This criterion highlights the multidimensional approach to protecting healthcare staff by designing and implementing a structured workplace violence prevention program. The program emphasizes zero tolerance for violence by addressing and mitigating potential risks that could affect the safety of employees. This program, which is usually overseen by the human resources department, should define the types of possible violence, the reporting procedures, and the consequences for those who engage in workplace violence. This policy serves as a guide to staff and management in recognizing, reporting, and addressing workplace violence instances. To further strengthen these efforts, healthcare organizations should conduct training or awareness campaigns to educate and empower staff in detecting, reporting, and preventing workplace violence. Moreover, it is important to conduct a risk assessment as part of the prevention program to identify potential sources of violence, followed by developing and implementing actions and measures to prevent these risks and protect the staff. In addition, security measures may include using panic buttons that staff may use in violent cases to alert others, access controls, the presence of security personnel, and the use of monitored surveillance cameras. Lastly, monitoring reported incidents related to workplace violence and taking appropriate actions to prevent reoccurrences are important to create a safe zone for the staff.*	
Evidence:	
4.2.1	*A written workplace violence prevention policy that outlines types of violence, reporting procedures, and workplace violence consequences.*
4.2.2	*Records of training or awareness campaigns for the staff on detection, reporting, and prevention of workplace violence.*
4.2.3	*Evidence of conducting a risk assessment to identify potential sources of violence, with action plans to prevent or mitigate them.*
4.2.4	*Evidence of implementing security measures to protect the staff, such as panic buttons, access controls, security staff, and surveillance cameras.*
4.2.5	*Evidence of quarterly reviewing and monitoring reported incidents related to workplace violence, with actions taken to prevent reoccurrences.*

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Criteria 4.3: the organization implements an occupational health and safety (OHS) program to promote a healthy work environment.	
Description: *This criterion underlines the importance of establishing a comprehensive occupational health and safety (OHS) program with the primary objective of fostering a healthy work environment and protecting staff's physical and psychological well-being. This program is a roadmap for hazard identification, risk assessment, and mitigation strategies. To promote the program's effectiveness, healthcare organizations should train the staff on relevant occupational health and workplace safety topics to equip them with the knowledge and skills required to identify and reduce the risk of workplace incidents. The OHS program encompasses multifaceted initiatives, including but not limited to risk assessments, pre-employment assessments, vaccinations, safety training, and health and medical surveillance. For instance, conducting regular proactive workplace risk assessments to identify potential occupational hazards followed by action plans and measures (i.e. corrective and preventative) to prevent or mitigate these risks would protect the health and safety of staff. Lastly, healthcare organizations must monitor the reported and investigated workplace incidents. Such monitoring could be through performance indicators, tracking records, root cause analysis reports, or staff feedback analysis. Compliance with these requirements is a fundamental commitment to the safety of healthcare workers and patients.*	
Evidence:	
4.3.1	*A written occupational health and safety (OHS) program that outlines the strategies for preventing and managing occupational health hazards.*
4.3.2	*Records of staff training on relevant occupational health and workplace safety topics.*
4.3.3	*Evidence of integrating and implementing pre-employment health screening and vaccinations as part of the OHS program.*
4.3.4	*Evidence of conducting a workplace risk assessment, at least annually, to identify occupational hazards and take actions to prevent them.*
4.3.5	*Evidence of reviewing and monitoring reported, investigated and managed workplace incidents and injuries.*

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Criteria 4.4: the organization introduced a work-life balance program to support staff members' physical and psychological well-being.	
Description: *This criterion highlights the impact of a work-life balance program on fostering a healthy and sustainable work environment. The program aims to support staff in achieving a harmonious balance between their professional responsibilities and personal lives, ensuring their physical and psychological well-being. Subsequently, it enhances staff satisfaction, reduces burnout, improves retention rates, and ultimately contributes to better patient outcomes. However, to ensure its best utilization, healthcare organizations need to communicate the program and its benefits with all staff through educational materials, announcements, posters, emails, newsletters, or any other channel to ensure that staff are aware of available work-life balance resources. The work-life balance program may include various staff wellness initiatives such as flexible work assignments, remote work policy, stress management workshops, mental health support resources, healthy eating campaigns, meditation sessions, and fitness programs. Also, the program includes implementing time-off policies, including vacation, sick, and parental leave. These initiatives help staff manage the unique stresses of healthcare and contribute to better physical and mental health. Lastly, healthcare organizations must continually review and monitor the effectiveness of the work-life balance program and assess the rate of staff participating to gauge its impact and make necessary adjustments.*	
Evidence:	
4.4.1	*A written work-life balance program outlines the objectives, components, scope, procedures, and responsibilities.*
4.4.2	*Evidence of using communication and education materials to inform staff members about the work-life balance program.*
4.4.3	*Evidence of implementing staff wellness initiatives such as flexible work assignments, stress management workshops, and fitness programs.*
4.4.4	*Evidence of implementing time-off policies to regulate and manage staff leaves, including vacation, sick, and parental leave.*
4.4.5	*Evidence of reviewing and monitoring the effectiveness and participation rates in the work-life balance program at least quarterly.*

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Criteria 4.5: the organization adopts the Practice Partnership Model of Care (PPM) to achieve better patient and workforce outcomes.	
Description: *This criterion highlights the effect of the Practice Partnership Model (PPM) on patient and staff outcomes. The model emphasizes partnership among healthcare providers. It includes four main components: working in partnership with other staff; clinical handover at the bedside to promote greater patient involvement in their care; comfort rounds (i.e. conducting regular care rounds every 1-2 hours to address patient needs immediately); and environmental modifications (i.e. increasing the time that clinicians spend in the direct vicinity of their patients). These four components reflect patient-centeredness, transform care at the bedside, and improve patient and staff outcomes. Effective PPM implementation starts with educating clinical staff about PPM principles and practices. Such education equips healthcare professionals with the necessary knowledge and skills to effectively implement the model and foster a collaborative care approach. Furthermore, health organizations should integrate PPM principles into the clinical handoff process to enhance communication and cooperation during these critical transitions. Also, documenting the integration of PPM into patient care processes in the patient medical record (e.g. in the multidisciplinary care plan) is essential, as it ensures that the patient's care plan aligns with the collaborative approach. Lastly, healthcare organizations should monitor the impact of PPM on patient and workforce outcomes, such as patient satisfaction scores, staff satisfaction surveys, and clinical outcome measures.*	
Evidence:	
4.5.1	*A written policy or guideline outlines the Practice Partnership Model (PPM) principles, objectives, and implementation strategies.*
4.5.2	*Records of training or orientation sessions designed to familiarize clinical staff with PPM principles and practices.*
4.5.3	*Evidence of integrating the PPM principles in the clinical handoff process across different departments or units.*
4.5.4	*Evidence of documenting the integration of PPM into patient care processes in the patient medical record.*
4.5.5	*Evidence of reviewing and monitoring the impact of PPM on patient and workforce outcomes, such as patient and staff satisfaction.*

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Domain 5: Patient Safety Training

Patient safety is a cornerstone of high-quality healthcare, and its significance cannot be overstated in ensuring positive health outcomes for patients. To achieve optimal and safe patient care, healthcare organizations must prioritize patient safety training and education for their staff [49]. In fact, patient safety training equips healthcare professionals with the knowledge and skills necessary to identify potential risks, prevent errors, and enhance patient care [50]. Hence, comprehensive patient safety training empowers staff members to adopt evidence-based practices, communication techniques, and teamwork strategies that foster a safety culture.

The journey of patient safety training starts upon hiring new staff members through integrating patient safety into general orientation programs [51]. Patient safety terminologies and other topics such as error reporting and just culture are discussed during orientation to set the foundation for a safety-oriented work environment [52]. However, due to the progressive nature of patient safety practices, conducting a follow-up annual patient safety training is vital to reinforce patient safety principles consistently [53]. Through regular training sessions, staff members stay updated on the latest protocols, guidelines, and advancements in patient safety, leading to improved patient outcomes and minimized medical errors [54]. Annual training ensures that all staff members remain well-versed in patient safety concepts and continue to prioritize safe and high-quality care delivery. On top of that, the role of simulation-based training appears a promising effect in supporting patient safety. Simulation-based training offers a dynamic and realistic learning environment where clinical staff can practice and refine their skills without jeopardizing patient safety [55]. Through simulated scenarios, healthcare professionals encounter lifelike patient care challenges, develop crisis management skills, and enhance teamwork and communication. This subsequently prepares staff to respond effectively to complex situations and ultimately improve patient safety [56].

Several factors affect the successful implementation of patient safety training during the orientation phase and annually. Having leaders certified in patient safety and qualified in healthcare leadership appears to be among the most important affecting factors [57]. The significance of having leaders certified in patient safety and qualified in healthcare leadership cannot be overstated. Leaders with expertise in patient safety play a pivotal role in steering patient safety initiatives, driving a proactive approach to risk management, and inspiring a safety-oriented culture among their teams [58]. Certification and qualification ensure that leaders are equipped to make informed decisions, mentor their teams, and effectively advocate for patient safety at all levels of the organization [59].

In conclusion, patient safety training empowers clinical staff with the knowledge and skills to provide safe and effective patient care. The commitment to ongoing education ensures that healthcare professionals remain vigilant in pursuing patient safety excellence, resulting in improved patient outcomes and enhanced trust in the healthcare system.

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Criteria 5.1: the organization integrates patient safety terminologies and practices in the general orientation program for all new hires.	
Description: *This criterion emphasizes the importance of incorporating patient safety terminologies and practices into the orientation process of all newly recruited staff. The goal is to familiarize them with essential safety terminologies, including patient safety goals, just culture, systems thinking, patient engagement, adverse events, near misses, error reporting, and root cause analysis. Hence, healthcare organizations are required to have a written orientation policy that clearly defines patient safety content included in the orientation program to provide a structured framework for educating new staff members on patient safety fundamentals. For consistency, a written agenda, curriculum, or training materials are also vital to equip new hires with a consistent understanding of patient safety terminologies and practices. Following these orientation programs, organizations are expected to maintain orientation delivery, attendance, and evaluation records. Further, to ensure that orientation sessions are delivered correctly and using updated patient safety information, healthcare organizations must assign the orientation role to trainers qualified or experienced in patient safety. Lastly, reviewing and monitoring the enrollment rate of new hires in the orientation program is a critical indicator to ensure that the vast majority of new staff members receive this vital orientation promptly to prevent or reduce adverse events and enhance the overall patient safety.*	
Evidence:	
5.1.1	*A written orientation policy integrates patient safety terminologies and practices as an integral part of the organizational orientation program.*
5.1.2	*A written agenda, curriculum, or training materials that integrate patient safety terminologies and practices in the orientation program.*
5.1.3	*Evidence of delivering orientation programs that include patient safety practices by trainers qualified or experienced in patient safety.*
5.1.4	*Evidence of performing a post-orientation evaluation to assess the effectiveness of the orientation program.*
5.1.5	*Evidence of reviewing and monitoring the enrollment rate of the new hires in the orientation program on an annual base.*

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Criteria 5.2: the organization provides annual patient safety training to staff members to promote adherence to patient safety practices.	
Description: *This criterion emphasizes the necessity of annual updated training on various patient safety subjects for all staff members based on the organization's needs and priorities. The training aims to equip healthcare professionals with the necessary knowledge and skills to improve patient outcomes and reduce adverse events. The annual patient safety training is different from the general patient safety orientation as it is longer in time, has more comprehensive content, targets new and previously hired staff, and aims to reinforce and update the knowledge and skills of healthcare staff regarding patient safety. This training may cover patient safety goals, error prevention and reporting, communication strategies, infection control, and medication safety. Hence, healthcare organizations are expected to establish a written policy, plan, or guideline that outlines the content, frequency, documentation, and responsibility for this training. Additionally, having a written agenda, curriculum, and schedule demonstrates training regularity and consistency. Following this training, organizations are expected to maintain orientation delivery, attendance, and evaluation records. Also, these records should be used in the staff performance evaluations and to enhance the program content for future use. Finally, reviewing and monitoring the enrollment rate of staff in the annual patient safety training is an important indicator to ensure that a substantial percentage of the workforce is regularly exposed to patient safety education.*	
Evidence:	
5.2.1	*A written policy, plan, or guideline that defines the content, frequency, documentation, and responsibility of the annual patient safety training.*
5.2.2	*Records or attendance reports demonstrate the regular delivery of patient safety training for all staff.*
5.2.3	*Evidence of integrating the annual patient safety training in the annual performance evaluations of all staff members.*
5.2.4	*Evidence of performing a post-training assessment or evaluation to assess the training effectiveness, with training evidence kept in staff personal files.*
5.2.5	*Evidence of reviewing and monitoring the enrollment rate of staff in the annual patient safety training.*

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Criteria 5.3: the organization provides staff with training in teamwork, using an evidence-based teamwork tool, to enhance patient safety.	
Description: *This criterion highlights the importance of promoting effective teamwork among healthcare staff. Teamwork training fosters a collaborative culture that reduces medical errors and enhances patient outcomes. In this, healthcare organizations must integrate teamwork training in their training plans. This integration shall be accompanied and supported by adopting an evidence-based teamwork tool such as Crew Resource Management (CRM) or Team Strategies and Tools to Enhance Performance and Patient Safety (TeamSTEPPS®). CRM and TeamSTEPPS are evidence-based teamwork tools to optimize patient outcomes by improving teamwork skills among healthcare teams. As part of this training, organizations need to ensure that staff members (i.e. clinical and non-clinical) are equipped with the knowledge and skills required for successful team collaboration, teamwork, and conflict resolution. Including clinical and non-clinical staff in this training demonstrates a commitment to enhancing teamwork and ensures that a broad spectrum of staff is prepared to engage in effective teamwork. Lastly, organizations need to review and monitor the implementation of teamwork skills within the organization, as well as the impact of teamwork training on patient safety. Such monitoring could be through performance indicators, audit rounds, leadership rounds, executive committee discussions, or staff feedback reports.*	
Evidence:	
5.3.1	*Evidence indicates that teamwork training, including teamwork skills and conflict resolution, should be integrated into the organizational training plan.*
5.3.2	*Evidence indicates adopting an evidence-based teamwork tool to train the staff on the principles of effective teamwork.*
5.3.3	*Documented evidence represents the participation of clinical staff in teamwork training programs.*
5.3.4	*Documented evidence represents the participation of non-clinical staff in teamwork training programs.*
5.3.5	*Evidence of reviewing and monitoring the implementation of teamwork skills in the organization and how it impacts patient safety.*

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Criteria 5.4: the organization maintains a sufficient number of leaders and managers certified or trained in advanced patient safety.	
Description: *This criterion underscores the vital need for a sufficient number of leaders and managers certified or trained in advanced patient safety practices. Such training levels shape the culture of patient safety within healthcare organizations and equip leaders with deep knowledge and understanding of patient safety principles, error prevention strategies, and risk management. Subsequently, the training assists them in effectively leading patient safety initiatives and empowers them to make informed decisions regarding patient safety priorities. In this, healthcare organizations are expected to ensure that a sufficient number of organizational leaders and managers (i.e. 10% or more based on the organization's size, scope, and structure) have attended advanced safety training as determined by the healthcare organization. To foster the concept of advanced patient safety training, healthcare organizations need to integrate this training into the job descriptions and the performance evaluations of leaders and managers. This would emphasize the importance of patient safety as a core competency for these roles and setting clear expectations. Lastly, it is important to monitor the impact of trained leaders and managers on patient safety outcomes, such as improved reporting of incidents, improved rate of closed incidents, improved hand hygiene, and decreased error rate. These numbers provide concrete evidence of the impact of these training initiatives and subsequently enhancing patient safety.*	
Evidence:	
5.4.1	*Evidence indicates that a representative number of leaders attended advanced safety training based on the organization's size and structure.*
5.4.2	*Evidence indicates that a representative number of managers attended advanced safety training based on the organization's size and structure.*
5.4.3	*Evidence of integrating patient safety training requirements in the job descriptions for leaders and managers.*
5.4.4	*Evidence of integrating patient safety training in the performance evaluations of leaders and managers.*
5.4.5	*Evidence of reviewing and monitoring the impact of trained leaders and managers on patient safety outcomes (e.g. improved reporting).*

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Criteria 5.5: the organization provides simulation-based training for clinical staff to improve patient safety and reduce errors.	
Description: *This criterion highlights the importance of simulation-based training as a method to improve patient safety. This method allows clinical staff to practice skills in a realistic, risk-free environment, learn from mistakes, and develop teamwork skills. Simulation-based training can be used to train for various patient safety-related scenarios, such as emergency response, surgical simulations, medication administration scenarios, and communication simulations. Healthcare organizations need to establish a policy that outlines the indications, frequency, and scenarios for simulation-based training to provide clear guidance on when and how to employ this powerful training method. Therefore, the availability and accessibility of such simulation facilities and equipment are imperative and allow healthcare professionals to practice and refine their skills in a controlled environment. Following the simulation-based training, organizations are expected to keep training and attendance records for that. Moreover, evidence indicating leadership support for simulation-based training underscores its organizational commitment to patient safety improvement. Finally, the monitoring and reviewing improvements in safety indicators resulting from simulation training provide tangible evidence of its impact, such as improved post-resuscitation survival rate following life support simulation-based training or decreased surgical site infection following laparoscopy simulation-based training.*	
Evidence:	
5.5.1	*A written policy outlines the indications, process, frequency, and scenarios of simulation-based training.*
5.5.2	*Evidence that the simulation facilities and equipment are available and accessible to clinical staff.*
5.5.3	*Documented evidence of the participation of clinical staff in safety-related simulation-based training.*
5.5.4	*Evidence suggests that leadership supports the use of simulation-based training to improve safety.*
5.5.5	*Evidence of reviewing and monitoring improvements in safety indicators resulting from simulation training.*

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Domain 6: Patient and Family Engagement

Patient and family engagement is an evolving concept of modern healthcare that empowers patients to actively participate in their care journey [60]. This engagement is an ethical necessity that may reduce the likelihood of medical errors and adverse events [61]. Hence, healthcare organizations are encouraged to integrate patient and family engagement as a strategic priority to ensure their partnership in the care journey rather than being passive recipients of medical care. Patients and their families possess unique insights into their health conditions, preferences, and goals, making their active participation in decision-making crucial to achieving positive outcomes [62].

The literature revealed that patient and family engagement assists in improving patient adherence to treatment plans, enhancing patient safety and outcomes, increasing patient satisfaction, and reducing healthcare costs [63]. However, it should be acknowledged that there are hindrances to this engagement, including communication gaps, cultural barriers, health literacy disparities, and resistance from healthcare providers [64]. Therefore, engagement is not a simple outcome that can be achieved by implementing a single initiative.

Healthcare organizations adopt multifaceted models that underpin patient and family engagement, such as the health belief model, transtheoretical model, and shared decision-making model [65]. The effect of these models relies heavily on translating them into multiple initiatives to support patient engagement. These initiatives may include initiating patient education programs, establishing communication channels, creating shared decision-making practices, and establishing a patient and family advisory council (PFAC) [66]. For instance, educational programs can empower patients and their families with essential health information, helping them better understand their conditions and the treatment options that subsequently assist them in shared decision-making.

In addition to shared decision-making, organizations may opt to implement other approaches to further enhance the engagement of patients and their families. Hence, organizations gather, evaluate, and use data on satisfaction and complaints to promote patient safety by identifying areas for improvement based on patient feedback [67]. By leveraging this feedback, healthcare organizations can implement targeted interventions to enhance patient safety and overall care quality. Furthermore, timely and transparent disclosure of harmful adverse events occurring during care provision demonstrates accountability that can foster patient engagement and enhance commitment to learning from mistakes [68]. Finally, the establishment of a PFAC provides a platform for patients and their families to offer their perspectives and suggestions in an environment that truly values patient and family voices in shaping the delivery of healthcare services [69].

In conclusion, “patient and family engagement” is a vital component of patient safety. Integrating the engagement as a strategic priority and supporting this priority with various initiatives assist healthcare organizations in tapping into patients' and families' unique perspectives and expertise, resulting in better decision-making, improved care outcomes, and enhanced patient satisfaction.

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Criteria 6.1: the organization integrates patient and family engagement as a strategic priority supported by multiple initiatives.	
Description: *This criterion underscores the importance of patient engagement. Person-centered care and patient engagement are related concepts. However, person-centered care emphasizes tailoring healthcare services and treatments to individual needs, while engagement refers to the active involvement of patients in their healthcare journey. In this, the healthcare organization needs to explicitly integrate patient and family engagement as a central organizational priority. Such integration guides the active involvement of patients and their families in the care processes to empower them and improve overall health outcomes. To support this, organizations need to train staff members on patient engagement to ensure that healthcare professionals have the knowledge and skills to effectively engage patients and their families in the decision-making process. Further, organizations need to assess the level of patient engagement annually using validated tools such as WeCares or Patient Activation Measure (PAM) to address patient needs. The assessment shall be followed by implementing engagement initiatives such as Ask Me 3, SpeakUp, or the What Matters To You (WMTY) campaign to foster a collaborative relationship with healthcare providers. Monitoring the impact of these initiatives on improvements in patient clinical outcomes, patient satisfaction, complaint rate, and overall quality is important. Such monitoring could be through a set of performance indicators, inspection reports, staff feedback reports, or customer feedback reports.*	
Evidence:	
6.1.1	*A written strategic plan that integrates patient and family engagement as an organizational strategic priority.*
6.1.2	*Records of training or awareness campaigns for the staff on the importance and methods of patient and family engagement.*
6.1.3	*Evidence of annual assessment for patient engagement using an evidence-based validated tool such as WeCares or Patient Activation Measure (PAM).*
6.1.4	*Evidence of implementing patient and family engagement initiatives such as Ask Me 3©, SpeakUp©or What Matters To You (WMTY) campaign.*
6.1.5	*Evidence of reviewing and monitoring the impact of implementing patient and family engagement initiatives.*

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Criteria 6.2: the organization implements mechanisms to gather, review, and utilize patient and family satisfaction and complaint data.	
Description: *This criterion underscores the necessity of establishing robust mechanisms to collect, review, and learn from patient and family satisfaction and complaint data. These mechanisms provide valuable insights into the quality of care, services, and overall patient experience. Further, the organization gains a deeper understanding of risks and areas that require improvement and subsequently develops strategies to address concerns aligned with patient preferences and needs. In this, healthcare organizations need to establish a written policy that clearly outlines the processes and responsibilities related to collecting and analyzing patient and family satisfaction and complaint data. Additionally, using a standardized structured platform or tool (i.e. electronic or paper-based) to collect and analyze patient satisfaction and complaint data regularly. This approach helps in identifying patterns and trends in patient and family satisfaction and complaints, and this analysis should be followed by an action plan to address issues or areas for improvement identified. For transparency and accountability purposes, the satisfaction and complaint results shall be disseminated to involved stakeholders along with the corresponding action plan. Thereafter, the organization needs to monitor action plans and improvement initiatives to address issues identified in satisfaction and complaint data. This monitoring could be through a set of performance indicators, dashboards, or customer feedback reports.*	
Evidence:	
6.2.1	*A written policy outlines the processes, procedures, and responsibilities in collecting and analyzing patient & family satisfaction and complaint data.*
6.2.2	*A standardized structured platform or tool is available to collect and analyze patient and family satisfaction and complaint data regularly.*
6.2.3	*Evidence of collecting and analyzing patient and family satisfaction trends and complaint data, followed by a written action plan.*
6.2.4	*Documented evidence of disseminating patient and family satisfaction, complaint results, and action plans to relevant stakeholders.*
6.2.5	*Evidence of reviewing and monitoring action plans and improvement initiatives addressing identified issues in satisfaction and complaint data.*

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Criteria 6.3: the organization engages patients and their families in the decision-making regarding their treatment and care plans.	
Description: *This criterion emphasizes the commitment to patient-centered care and respect by actively engaging patients and their families in decision-making. Patient education and shared decision-making are related concepts. However, education provides patients with information to understand their health condition and treatment options, where information typically flows from the healthcare provider, while shared decision-making is a collaborative approach between patients and healthcare providers to make decisions, with information flows in both directions to arrive at a decision that is aligned with the patient's values, goals, and preferences. In this, health organizations need to have a policy outlining the process and structure of patient engagement and need to provide patients and their families with educational materials to assist them in making informed decisions. Further, organizations need to keep their staff aware of shared decision-making topics such as health literacy to facilitate meaningful engagement. Documenting shared decision-making related to treatment options, care plans, discharge planning, and discharge decisions in the patient's medical record is critical. Also, monitoring the impact of shared decision-making on clinical outcomes, patient satisfaction, and adherence to treatment plans is important. Such monitoring could be through a set of performance indicators, chart reviews, staff feedback reports, or customer feedback reports.*	
Evidence:	
6.3.1	*A written policy outlines the process, procedure, and responsibility for engaging patients and their families in treatment and care decisions.*
6.3.2	*Evidence of equipping patients and their families with educational materials that assist them in shared decision-making.*
6.3.3	*Documented evidence demonstrates shared decision-making in treatment and care plans, including informed consent.*
6.3.4	*Documented evidence demonstrates shared decision-making in discharge planning and discharge decisions.*
6.3.5	*Evidence of reviewing and monitoring the impact of shared decision-making on clinical outcomes, satisfaction, or adherence to treatment plans.*

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Criteria 6.4: the organization timely and openly discloses harmful adverse events occurring during care provision to the affected patient.

Description:

This criterion emphasizes the importance of informing patients of adverse events that may have occurred during their care in a transparent, timely, and factual manner. Such disclosure fosters trust, patient safety, accountability, and dedication to patient-centered care and ethical responsibility. Further, it enables patients to be informed about their care, facilitates an open dialogue between patients and healthcare providers, and supports a culture of learning and continuous improvement. In this, healthcare organizations need to establish a policy that outlines the process, timing, content, and documentation of disclosing harmful adverse events to patients. This disclosure must be conducted by an individual trained in disclosure processes to consider and accommodate the emotional and psychological needs of patients and their families. The details of the disclosure process must be documented, including a record of disclosure content, timing, the disclosing individual, and patient responses. Beyond disclosure, healthcare organizations need also to develop a post-disclosure analysis to prevent future reoccurrence. Lastly, monitoring compliance with the event disclosure policy and action plans is vital. Such monitoring could be through a set of performance indicators, chart reviews, root cause analysis reports, or customer feedback reports. These requirements prioritize patients' rights to know about adverse events and contribute to overall healthcare safety by reducing the likelihood of similar events in the future.

Evidence:

6.4.1	*A written policy outlines the process, procedure, timeframe, and responsibility of disclosing harmful adverse events to patients.*
6.4.2	*Records of training for individuals responsible for event disclosure on what, when, and how to effectively and timely disclose adverse events.*
6.4.3	*Documented evidence or records of events disclosed to affected patients, including disclosure content, time, disclosing individual, and response.*
6.4.4	*Documented evidence of developing post-disclosure analysis to document root causes, identified gaps, and actions to prevent future reoccurrence.*
6.4.5	*Evidence of reviewing and monitoring compliance with the event disclosure policy and the developed action plans at least quarterly.*

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Criteria 6.5: the organization establishes a patient and family advisory council (PFAC) to enhance patient-centered care and patient safety.	
Description: *This criterion emphasizes the crucial role of the Patient and Family Advisory Council (PFAC) in promoting patient-centered care, enhancing the patient experience, and ensuring patient safety. The PFAC comprises patients, family members, caregivers, and healthcare professionals collaborating to provide valuable insights and feedback, ensuring that health services align with patient needs, values, goals, and preferences. Hence, healthcare organizations need to establish a structured PFAC, with written terms of reference to define the council's charges and formation order to specify involved members. The membership list must include a patient or patient representative and must be led by the organization's director or a designated leader. This leadership is crucial in effectively guiding the council's activities and aligning them with the organization's goals. Moreover, maintaining records of regular PFAC meetings, focusing on achieving at least 70% closure of discussed agenda items, demonstrates the council's commitment to making tangible progress in enhancing patient-centered care. Finally, monitoring the implementation of action plans developed based on patient feedback within the PFAC reinforces the organization's dedication to translating patient input into meaningful improvements in care delivery. Such monitoring could be through a set of performance indicators, reviewing meeting minutes, tracking action plans, or analyzing patient feedback reports.*	
Evidence:	
6.5.1	*A written terms of reference defines the charges, roles, and responsibilities of the Patient and Family Advisory Council (PFAC).*
6.5.2	*A written membership list indicates that the organization's director or a designated leader is leading the PFAC.*
6.5.3	*Records of at least a quarterly meeting by PFAC, with at least 70% closure of the discussed agenda items.*
6.5.4	*Evidence of involving patients and family members in the discussion, decision-making, and co-development of policies in PFAC meetings.*
6.5.5	*Evidence of reviewing and monitoring the implementation of action plans that were developed based on patient feedback in PFAC at least quarterly.*

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Domain 7: Patient Safety Indicators

Patient safety indicators are paramount as they provide a quantitative framework to assess and enhance patient safety, prevent adverse events, and optimize patient outcomes [70]. By meticulously tracking and analyzing safety-related data, these indicators offer a real-time snapshot of an organization's performance in ensuring patient safety [71]. Linking patient safety indicators with strategic objectives establishes a symbiotic relationship where the pursuit of safety aligns with broader organizational goals [72]. In fact, patient safety indicators serve as a critical cornerstone of the organization's continuous quality improvement. At the heart of this initiative lies the meticulous collection of reliable data, accurately representing safety events and trends across care delivery [73]. The organization employs rigorous methodologies to gather and document data, ensuring its integrity and fidelity.

Once data is collected, a rigorous analysis ensues, unveiling patterns, outliers, and potential areas of concern [74]. Importantly, the organization adopts a multifaceted approach to validate the accuracy and consistency of the collected data [75]. By employing various validation techniques, such as cross-referencing with independent sources and conducting periodic audits, the organization guarantees the reliability of its patient safety indicators [76].

The journey does not end with data collection and validation. The organization conscientiously evaluates the results against predefined benchmarks or internal targets, striving for excellence and optimal patient safety outcomes [77]. This diligent scrutiny provides a holistic perspective, identifying both areas of triumph and those necessitating improvement.

What truly distinguishes this approach is the organization's steadfast commitment to a data-driven approach. Harnessing the insights gleaned from patient safety indicators, the organization systematically addresses areas of concern, embarking on evidence-based interventions and transformative actions [78]. This data-driven approach engenders a culture of proactive problem-solving and continuous learning, bolstering the organization's ability to preemptively mitigate risks and refine patient care protocols [79].

In summation, patient safety indicators constitute a dynamic and iterative process that underscores the organization's unwavering dedication to ensuring the highest patient safety and quality of care standards. Through rigorous data collection, meticulous analysis, robust validation techniques, benchmark evaluations, and data-driven interventions, the organization crafts a roadmap that identifies safety concerns and drives proactive, evidence-based enhancements. This commitment resonates throughout the organization, from leadership to frontline staff, propelling healthcare quality to unprecedented heights and ensuring patient safety remains at the heart of every endeavor.

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Criteria 7.1: the organization establishes a robust system for collecting reliable data related to patient safety indicators.	
Description: *This criterion emphasizes the importance of establishing a structured data collection system (i.e. electronic or paper-based) focused on gathering reliable and accurate data pertinent to patient safety indicators to ensure consistent data collection and allow for accurate assessments of patient safety performance. The ultimate goal of patient safety data collection is facilitating evidence-based decisions, early detection of potential concerns, and continual improvement in patient safety outcomes. Hence, healthcare organizations need to establish a written protocol to guide the process of collecting patient safety data. This protocol shall include a list of defined organizational-level patient safety indicators, including the structure, process, and outcome safety indicators. In each indicator, the operational definition (i.e. specific description of how a particular variable or data element is measured or observed) of each concept in the data collection tool is equally important to ensure consistency, accuracy, and reliability in data collection by the data custodians (i.e. individual responsible of collecting the data). Importantly, healthcare organizations need to undertake measures to secure the confidentiality of collected patient safety data for legal, ethical, and trust purposes. Lastly, organizations shall regularly monitor adherence to the data collection process, timeline, and frequency. Such monitoring could be through audit rounds, staff feedback reports, or a review of the process by relevant leaders, departments, or committees.*	
Evidence:	
7.1.1	*A written policy, guide, or protocol outlines the process, procedure, timeframe, and responsibilities of collecting patient safety data.*
7.1.2	*Evidence of using standardized and operationally defined data collection tools (i.e. paper-based or electronic) in collecting patient safety data.*
7.1.3	*Records demonstrate collected patient safety data by data custodians following a pre-defined process, timeline, and frequency.*
7.1.4	*Evidence demonstrates implementing measures to secure and protect the confidentiality of collected patient safety data.*
7.1.5	*Evidence of reviewing and monitoring adherence to the data collection process by the relevant leaders, departments, or committees.*

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Criteria 7.2: the organization analyzes collected data related to patient safety indicators to detect trends, patterns, and areas of concern.

Description:

This criterion underscores the necessity of establishing a process to systematically analyze and interpret the gathered data concerning patient safety indicators, aiming to detect notable trends, patterns, and areas that may require attention. In this, healthcare organizations need to establish a written protocol to guide the data analysis process, specifying who is responsible and what method is employed. This protocol is essential for providing a consistent and standardized approach to data analysis. Healthcare organizations collect quantitative (e.g. numerical measurements) and qualitative (e.g., narratives, patient stories, or feedback) data. Hence, using relevant statistical techniques in analyzing quantitative and qualitative data is important. Also, analyzing the data by trained and competent individuals is equally important to ensure the reliability of the analysis. Examples of quantitative statistical analysis may include descriptive, inferential, and correlation analysis, while qualitative data may be analyzed using content, thematic, or narrative analysis. Further, healthcare organizations should employ visualization tools (e.g. bar charts, pie charts, run charts, scatter plots) relevant to the type of analyzed data to facilitate detecting patterns, trends, and potential safety issues. Lastly, monitoring the analysis process by relevant leaders, departments, or committees through reviewing reports, dashboards, and staff feedback may enhance consistency and standardization of the analysis process.

Evidence:

7.2.1	*A written protocol or guide outlines the process, procedure, timeframe, and responsibilities of data analysis for patient safety indicators.*
7.2.2	*Records of training or competency assessment on data analysis process for individuals involved in analyzing patient safety data.*
7.2.3	*Evidence shows the use of relevant statistical techniques in analyzing quantitative and qualitative data for patient safety indicators.*
7.2.4	*Evidence shows the use of relevant visualization tools to detect trends, patterns, and areas of concern in patient safety indicators.*
7.2.5	*Evidence of reviewing and monitoring the data analysis process by the relevant leaders, departments, or committees.*

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Criteria 7.3: the organization employs various validation techniques to ensure the reliability of the data related to patient safety indicators.	
Description: *This criterion highlights the importance of the systematic use of a range of data validation techniques, such as cross-referencing, recalculation, sampling, reconciliation, audit trails, consistency checks, and expert review, to ensure and maintain the utmost reliability, integrity, trustworthiness, and authenticity of patient safety data that is used in patient safety indicators. This validation enables informed decision-making, accurate assessments of patient safety performance, and the identification of potential patient safety concerns. Hence, healthcare organizations need to establish a protocol that explicitly outlines what patient safety data needs to be validated, who is responsible for that, how it should be conducted, what validation techniques are to be used (i.e. at least one method), and the frequency at which it must occur. Such protocol serves as a blueprint to ensure consistency and integrity of patient safety data. In parallel, it is important to maintain records of training or competency assessments, for individuals involved in the data validation process must be trained to do so. Further, organizations must take action promptly to resolve errors or discrepancies identified during the validation process to ensure that inaccuracies do not compromise patient care. Finally, healthcare organizations need to review and monitor the data validation process, which includes collecting feedback from data users to confirm the precision and utility of the data.*	
Evidence:	
7.3.1	*A written plan, procedure, or protocol outlining who, how, and how often patient safety data validation is conducted.*
7.3.2	*Records of training or competency assessment on data validation process for individuals involved in validating patient safety data.*
7.3.3	*Evidence demonstrates using at least one method to validate patient safety data, including cross-referencing, recalculation, or re-sampling.*
7.3.4	*Evidence indicates resolving errors or discrepancies identified during the validation of patient safety data.*
7.3.5	*Evidence of reviewing and monitoring the data validation process, including feedback collection from data users regarding data accuracy.*

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Criteria 7.4: the organization evaluates the results of patient safety indicators against predefined benchmarks or internal targets.	
Description: *This criterion emphasizes examining patient safety outcomes against predetermined national or international benchmarks or internally set targets. This comparison provides a tool to evaluate the effectiveness of organizational patient safety efforts and provides insights regarding areas of success and areas necessitating improvement. To this end, healthcare organizations need to create a protocol that outlines how these targets are established, considering a range of internal and external factors. Internal factors may include historical data, organizational mission, patient demographics, and available resources, while external factors may include regulatory requirements, community expectations, and market dynamics. Selecting a benchmark or setting a target is not the role of a specific individual in healthcare settings. Instead, the process owner of each patient safety indicator possesses the ultimate responsibility of setting targets that are rationalized, appropriate, and realistic, and selecting benchmarks that are sensitive to the patient population and risk-adjusted (i.e. making meaningful comparisons by considering the varying levels of risk associated with different healthcare organizations, such as the difference between acute care and long-term facilities). Lastly, an annual review of the appropriateness of benchmarks and targets is vital for maintaining relevance and alignment with the organization's evolving patient safety objectives and current performance.*	
Evidence:	
7.4.1	*A written plan, procedure, or protocol outlining how internal targets or benchmarks are established based on internal and external factors.*
7.4.2	*Evidence of involving the process owner of patient safety indicators in the selection process for internal targets or benchmarks.*
7.4.3	*Evidence of using pre-defined benchmarks or internal targets for patient safety indicators.*
7.4.4	*Evidence indicates that the selected internal targets for patient safety indicators are rationalized, and the selected benchmarks are risk-adjusted.*
7.4.5	*Evidence of periodically reviewing the appropriateness of benchmarks and targets based on performance and updating them as required.*

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Criteria 7.5: the organization adopts a data-driven approach to address areas of concern in patient safety indicators systematically.	
Description: *This criterion signifies the role of a data-driven approach in conducting improvement and addressing identified areas of concern using patient safety indicators. To begin, healthcare organizations need to integrate areas of improvement identified through patient safety indicators in a written patient safety plan to provide a structured framework for translating data into meaningful action. Equally important is employing a structured method such as performance indicator cards, dashboards, or balance scorecards to share the patient safety indicator results with relevant stakeholders. This approach empowers healthcare professionals, administrators, and patients to make informed choices that promote patient safety. Also, it fosters the use of data analysis in driving targeted interventions, timely adjustments, and a safer healthcare environment. Once an area of improvement is identified based on the results of patient safety indicators, healthcare organizations need to initiate performance improvement projects using systematic methodologies such as PDSA. In this, involving the right interdisciplinary team is important to ensure the effectiveness of performance improvement. Lastly, monitoring the progress in patient safety improvement initiatives and action plans by relevant leaders or committees is pivotal to ensure accountability and track the impact of these initiatives. Such monitoring could be through inspection reports, audit rounds, dashboards, and seeking feedback from patients and staff.*	
Evidence:	
7.5.1	*A written patient safety plan addresses areas of concern identified based on results from patient safety indicators.*
7.5.2	*Evidence of sharing the results of patient safety indicators with relevant stakeholders to ease data-driven decisions.*
7.5.3	*Evidence of establishing improvement initiatives or action plans based on the results of patient safety indicators.*
7.5.4	*Evidence of involving relevant interdisciplinary teams in patient safety data-driven decisions.*
7.5.5	*Evidence of reviewing and monitoring the progress and action plans by relevant leaders or committees at least quarterly.*

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Domain 8: Technology for Patient Safety

In healthcare, technological solutions refer to integrating advanced tools, systems, and equipment to optimize patient care, prevent errors, and enhance patient safety [80]. Examples of these technologies may include an electronic health record (EHR), clinical decision support systems, smart infusion pumps, automated hand hygiene monitoring, cardiac telemetry, wearable fall prevention sensors, and medical imaging technology.

The use of technology is important as it serves as a patient safety strategy. For instance, the infusion pump is a technological solution designed to deliver precise medication doses while minimizing the risk of adverse reactions [81]. Safe infusion pumps are pivotal in minimizing medication errors during intravenous drug administration. These pumps are equipped with smart features like drug libraries, dose calculations, and customizable alarm settings to ensure accurate dosing and contribute to improved safety outcomes [82]. Although the infusion pump is part of many other technological solutions, it gains high importance due to its link with immediate serious harm or sentinel events if not used properly or maintained effectively [83]. According to scientific data on sentinel events, a substantial proportion of these events involve medical devices, including infusion pumps. A guideline that was published by the American Society of Health-System Pharmacists (ASHP) has highlighted instances where infusion pump-related errors have led to patient harm [84].

The EHR system is another impactful patient safety application [85]. The system streamlines patient information, treatment plans, and medical histories, ensuring accurate and accessible data for healthcare providers. Such systems assist in significantly reducing errors due to incomplete records and allow making informed decisions about patient care [86]. Moreover, EHR systems facilitate seamless communication among care teams, supporting coordinated care across different departments and settings [87].

A wide array of other technological solutions supports patient safety. For instance, barcode scanning, automated dispensing cabinets, medication reconciliation software, and computerized physician order entry (CPOE) systems augment medication safety by enhancing prescription, dispensing, and administration accuracy, thereby reducing errors and adverse events [88]. It should be acknowledged, however, that despite the numerous benefits that technological solutions offer, these technologies can inadvertently lead to new types of challenges or errors such as alarm fatigue and infusion pump programming errors [89]. To mitigate these challenges, understanding technologies' limitations and commitment to ongoing training is crucial. For instance, the clinical alarm management system mitigates alarm fatigue and enables prompt responses to critical situations through prioritizing alerts, customizing alarm settings, and minimizing unnecessary alarms [90].

In conclusion, integrating the power of technology in healthcare organizations enhances patient safety outcomes, improves clinical workflows, and establishes an environment of comprehensive and safer patient care [91].

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Criteria 8.1: the organization employs a range of technological solutions to supplement and promote patient safety outcomes.	
Description: *This criterion highlights the healthcare organization's commitment to integrating an array of advanced technological solutions that synergistically enhance patient safety outcomes and reduce errors. Examples of these technologies may include clinical decision support systems, automated hand hygiene monitoring, cardiac telemetry, wearable fall prevention sensors, and medical imaging technology. Establishing written criteria that carefully consider patient safety features, user feedback, and a comprehensive patient safety risk assessment is a foundational step in this process. Patient safety features refer to a digital characteristic designed to minimize risks, prevent errors, and enhance safe patient care. Subsequently, conducting the required training or competency to relevant staff on these technologies is indispensable to enable them to operate and utilize the benefits of these tools effectively. To ensure these technological solutions are effective and user-friendly, end-user feedback reports are invaluable, providing insights into the practicality, efficiency, and usability. Lastly, the ongoing review and monitoring of the impact of these adopted solutions on patient safety outcomes allows healthcare organizations to maximize the use of these technologies and address identified gaps. Such monitoring could be through inspection reports, audit rounds, staff feedback reports, or customer feedback reports.*	
Evidence:	
8.1.1	*Written criteria consider patient safety features, users' feedback, and patient safety risk assessment when selecting technological solutions.*
8.1.2	*Records of staff training or competency assessment on the effective use of various technologies relevant to the involved staff.*
8.1.3	*Evidence of adopting at least five evidence-based technological solutions that promote safety.*
8.1.4	*Feedback report from users regarding the usability of the employed technological solutions.*
8.1.5	*Evidence of reviewing and monitoring the impact of adopted technological solutions on patient safety outcomes.*

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Criteria 8.2: the organization establishes and monitors a structured process to guide the safe and reliable use of infusion pumps.

Description:

This criterion underlines the importance of instituting a systematic approach to ensure the safety of infusion and syringe pumps since their use has become standard practice in healthcare to administer critical fluids to patients. Therefore, designing a system to assure proper pump selection, alarm management, programming, and staff training appears to be of tremendous benefit in enhancing the effective and safe use of infusion pumps, thereby improving patient safety and outcomes. Initially, healthcare organizations need to establish a policy that governs these pumps' use, maintenance, and monitoring. These pumps shall have safety features such as an anti-free flow system, air-in-line detection, and audible and visual alarms. These features may prevent medication errors, air embolisms, and other potential complications. Thereafter, training or competency on properly handling these pumps is essential to equip healthcare staff with the skills and knowledge required to operate the pumps accurately and effectively. To ensure these pumps' safe and precise operation, it is vital to regularly perform inspection, calibration, and maintenance by trained biomedical engineers. Lastly, reviewing and monitoring the compliance of healthcare staff with the infusion pump policy provides opportunities for ongoing improvement and optimization. Such monitoring may include periodic audits, checks, and feedback mechanisms.

Evidence:

8.2.1	*A written policy that outlines the use, maintenance, and monitoring of intravenous infusion pumps.*
8.2.2	*Records of training or competency assessment on the safe and effective handling of intravenous infusion pumps for staff using these pumps.*
8.2.3	*Evidence of using infusion pumps equipped with safety features, including anti-free flow, air-in-line detection, and audible and visual alarms.*
8.2.4	*A report indicates regular inspection, calibration, and maintenance of intravenous infusion pumps by trained biomedical engineers.*
8.2.5	*Evidence of reviewing and monitoring adherence to the policy of intravenous infusion pumps at least quarterly.*

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Criteria 8.3: the organization adopts and uses an electronic health record (EHR) system designed to enhance patient safety.	
Description: *This criterion underscores the strategic importance of adopting and employing an electronic health record (EHR) system in promoting patient safety and the overall quality of care. EHR is a cutting-edge technology that facilitates real-time information accessibility to enhance clinical decisions, reinforces patient safety practices, and improves documentation consistency. Initially, healthcare organizations need to select an EHR system based on patient safety features, user feedback, and a thorough patient safety risk assessment to ensure that the chosen system aligns with the organizational priorities. The effective use of EHR systems must be ensured through training for EHR users on various aspects such as data entry, data retrieval, and access permissions. This training aims to enhance staff competency and ensure they can utilize the EHR daily. From a patient safety perspective, it is important to incorporate patient safety standards in the HER, such as fall risk assessments, pressure injury risk assessments, and critical test result reporting. It is also important to have an EHR system with data retrieval capacity to enable healthcare organizations to retrieve patient safety data from the HER for improvement. Lastly, the regular review and monitoring of compliance with EHR clinical documentation and access privileges are essential for maintaining data accuracy, integrity, and adherence to established standards. Such monitoring could be through a set of performance indicators, audit rounds, chart reviews, or staff feedback reports.*	
Evidence:	
8.3.1	*Evidence of using an EHR system to enhance patient safety and reduce errors.*
8.3.2	*Records of staff training or competency assessment for EHR users on the effective use of the system, including accessing medical files and data entry.*
8.3.3	*Evidence of integrating patient safety practices and standards in the EHR system, such as fall risk assessment and critical test result reporting.*
8.3.4	*Evidence demonstrates that the EHR system is utilized as a source to retrieve patient safety data for improvement purposes.*
8.3.5	*Evidence of reviewing and monitoring compliance with the EHR clinical documentation and access privileges at least quarterly.*

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Criteria 8.4: the organization leverages technological solutions to enhance medication safety and reduce medication errors.	
Description: *This criterion emphasizes the role of technology in improving medication safety. Computerized Physician Order Entry (CPOE), Barcoding Medication Administration (BCMA), and medication-automated dispensing cabinets are examples of technologies that enable real-time medication verification, allergy cross-referencing, and streamlined communication among healthcare providers, and therefore improve medication safety and reduce medication errors. In this, healthcare organizations need to establish criteria for selecting technologies considering medication safety risk assessment and user feedback to ensure that the chosen technologies are needed, appropriate, and aligned with the organization's safety goals. Next, training healthcare staff to use these technological solutions is important to guarantee the effective use of these systems. Both CPOE and BCMA systems play complementary roles in enhancing medication safety and reducing medication errors. CPOE assists in ensuring that accurate and safe medication orders are placed at the prescription phase, while BCMA assists in verifying that the right medication is administered to the right patient at the right time, at the administration phase. Finally, reviewing and monitoring the impact of medication-related technologies on the rate of medication errors is a practice that allows making informed decisions on medication safety practices. Such monitoring could be through a set of performance indicators, audit rounds, chart reviews, or customer feedback reports.*	
Evidence:	
8.4.1	*Written criteria considering medication safety risk assessment and user feedback when selecting medication safety technologies.*
8.4.2	*Records of staff training or competency assessment on the effective use of medication-related technological solutions.*
8.4.3	*Evidence of adopting technologies such as Computerized Physician Order Entry (CPOE) to reduce medication prescription errors.*
8.4.4	*Evidence of adopting technologies such as Barcoding Medication Administration (BCMA) to reduce medication administration errors.*
8.4.5	*Evidence of reviewing and monitoring the impact of medication-related technologies on the rate of medication errors.*

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Criteria 8.5: the organization implements a clinical alarm management system to reduce equipment-related alarm fatigue and enhance safety.	
Description: *This criterion emphasizes the importance of implementing a clinical alarm management system to lessen alarm fatigue arising from medical equipment, thereby enhancing patient safety. Alarm fatigue can lead to missed alarms, which can have severe consequences for patients. Hence, ensuring that alarms are appropriately prioritized, tailored to patient needs, and seamlessly integrated with clinical workflows is vital. Healthcare organizations need to establish a policy that outlines the process of clinical alarm management, setting alarm parameters, and alarm escalation procedures. This policy shall incorporate all critical devices, including cardiac monitors, ventilators, defibrillators, infusion pumps, hemodialysis machines…etc. Equally important is the training and competency of nurses and technicians on the alarm system to allow effective and prompt responses to alarms. Further, organizations need to set alarm threshold settings customized based on the patient's clinical needs to minimize the risk of adverse events. Additionally, reporting and analyzing events or incidents related to clinical alarms is instrumental in identifying areas for improvement in the alarm management system. Lastly, at least monthly, reviewing and monitoring compliance with the clinical alarm system policy ensures that the policy remains implemented. Such monitoring could be through the reporting system, inspection reports, audit rounds, or staff feedback reports.*	
Evidence:	
8.5.1	*A written policy outlines the management of clinical alarms, including setting alarm parameters and alarm escalation procedures.*
8.5.2	*Records of training or competency assessment for nurses and technicians on the alarm system, its purpose, and how to respond effectively.*
8.5.3	*Documented evidence indicates customizing alarm threshold settings based on patient clinical needs.*
8.5.4	*Evidence indicates reporting and analyzing events or incidents related to clinical alarms using the reporting system, including frequent false alarms.*
8.5.5	*Evidence of reviewing and monitoring compliance with the clinical alarm system policy at least monthly, with a corrective action plan.*

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Domain 9: Evidence-based Safety Practices

In healthcare, the importance of patient safety prompts the adoption of evidence-based practices to prevent errors and adverse events. In recent years, mounting evidence has underscored the role of evidence-based practices in preventing adverse events and enhancing patient outcomes across a wide spectrum of healthcare domains [92]. Due to the large impact and frequency, safe medication administration is pivotal to patient safety. A written policy delineating medication preparation, administration, and dosage calculation lays the foundation for safe medication practices [93]. This policy standardizes medication administration practices and underscores the importance of standardizing the preparation and dilution of medications, particularly high-alert medications. Further, the employment of independent double-checking and standardized medication administration timing emerge as invaluable practices that act as a barrier against errors [94].

Safe blood transfusion is another evidence-based safety practice that stands as a cornerstone of patient care. Employing evidence-based blood transfusion and reaction management guidelines minimizes potential risks and optimizes patient outcomes [95]. In this, several practices are necessary to adopt, such as keeping the patient informed before blood transfusion, leveraging technology in the patient identification process before blood transfusion, using real-time monitoring during blood transfusions, and ensuring that staff members are trained and competent in the safe handling of blood and blood products [96].

Surgical and procedural safety is also vital to the patient safety framework. Hence, surgical safety protocols ensure safety through every stage of surgical or invasive procedures [97]. For instance, the utilization of the World Health Organization (WHO) surgical safety checklists, including pre-operative verification, site marking, and "time-out", protects patients against preventable errors [98]. Similarly, implementing multiple counts of surgical instruments, sharps, and sponges underscores the systematic approach to maintaining an accurate inventory throughout the procedure [99]. These practices mitigate potential surgical errors and enhance patient safety.

Despite their criticality, practices aimed at enhancing diagnostic accuracy occupy a less important position in the patient safety field. Hence, we highlight its importance here in this domain. Fighting against misdiagnosis is directly linked to better outcomes, and adopting various actions to prevent such errors is vital in completing the safety picture [100]. These actions include embracing advanced diagnostic technologies such as Clinical Decision Support Systems (CDSS), seeking timely consultations, using checklists and clinical pathways, and implementing a peer review system [101]. For instance, the strategic use of checklists, algorithms, and clinical pathways refines the diagnostic journey and guides medical evaluations with standardized and systematic approaches [102].

Last but not least, active engagement in patient safety research fuels the advancement of safety protocols and spearheads interventions that fundamentally enhance patient safety [103]. This research resonates beyond institutional boundaries, shaping healthcare policy and practice with empirical insights on a larger scale.

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Criteria 9.1: the organization implements evidence-based practices to ensure safe medication administration.	
Description: *This criterion underscores the importance of safe administration of medications. Healthcare organizations need to prioritize medication administration policies to guide the medication administration process, accurate dosage calculation, appropriate administration routes, and standardized dilution practices. To ensure the implementation of these policies, training of nursing staff on medication administration and dosage calculation appears to be highly valuable. In line with the latest medical research, minimizing the potential risk of medication errors and adverse events requires deploying a bundle of actions at all medication nodes. However, medication administration is often considered the most critical node that requires more attention due to the high volume, complexity, human factors, patient variability, and direct impact on patient safety. For instance, employing an independent double-check system before administering high-risk medications is an important safety measure to reduce medication errors. Further, adopting standardized medication administration timing would minimize variability and increase adherence to the prescribed regimens. Lastly, reviewing and monitoring medication administration errors by the concerned leaders, departments, and committees provide an opportunity to learn from mistakes and implement data-driven corrective measures accordingly. Such monitoring could be through a set of performance indicators, audit rounds, chart reviews, committees' discussions, or staff feedback reports.*	
Evidence:	
9.1.1	*A written policy outlines the medication administration process, including dosage calculation, administration routes, and standardized dilution.*
9.1.2	*Records of training or competency assessment for nurses on medication administration and dosage calculation.*
9.1.3	*Evidence of employing independent double-checking before administering high-risk medications.*
9.1.4	*Evidence of using a consistent and standardized medication administration timing across the organization.*
9.1.5	*Evidence of reviewing and monitoring medication administration errors, with an action plan developed accordingly.*

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Criteria 9.2: the organization implements evidence-based practices to administer blood and blood products safely.	
Description: *This criterion underscores the importance of safe blood transfusion practices. Blood administration safety protocol includes pre-administration assessments, accurate patient identification, meticulous cross-matching procedures, proper product storage and handling, real-time monitoring during transfusion, and adverse event reporting and analysis. This approach minimizes the potential for transfusion errors, allergic reactions, and adverse events. Therefore, healthcare organizations need to prioritize the establishment and adherence to blood transfusion policies that are meticulously aligned with evidence-based guidelines and best practices. To complement this, nursing staff must be trained and competent in blood administration, handling, and monitoring during transfusions. It should be acknowledged that preventing transfusion-related errors requires implementing several actions. For instance, employing an independent double-check before issuing and administering blood is a critical safety measure to reduce the potential for transfusion errors. Furthermore, a standardized real-time monitoring process during blood transfusions allows for immediate detection and response to any adverse events or reactions, thus enhancing patient safety. Lastly, reviewing and monitoring adverse transfusion events provides an opportunity to implement data-driven corrective measures accordingly. Such monitoring could be through a set of performance indicators, audit rounds, chart reviews, committees' discussions, or staff feedback reports.*	
Evidence:	
9.2.1	*A written policy outlines blood transfusion and reaction practices aligned with evidence-based guidelines and best practices.*
9.2.2	*Records of training or competency assessment for nurses on blood administration, handling, and monitoring.*
9.2.3	*Evidence of employing independent double-checking before issuing and before administering blood and blood products.*
9.2.4	*Evidence of using a standardized real-time monitoring process of patients during blood transfusions.*
9.2.5	*Evidence of reviewing and monitoring adverse transfusion events, with an action plan developed accordingly.*

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Criteria 9.3: the organization implements evidence-based practices to ensure surgical and procedural safety.	
Description: *This criterion emphasizes the importance of implementing evidence-based guidelines to reduce surgical errors. As the surgical environment presents additional safety challenges, creating a safety system that includes pre-operative assessments, surgical safety alerts, streamlined communication, and rigorous post-procedural care is crucial. Hence, healthcare organizations need to establish surgical and procedural safety policies to guide the safety aspects of surgical and invasive procedures. To reinforce this and to reduce surgical errors, healthcare organizations must implement a standardized surgical safety checklist that includes sign-in (i.e. before anesthesia induction and surgery begins), time-out (i.e. just before the surgical incision), and sign-out (i.e. before the patient leaves the operating or procedural room) phases. In this, the adoption of surgical site marking and "time-out" processes must be standardized for all surgical operations and selective high-risk invasive procedures (e.g. central line insertion, chest tube insertion, and lumber puncture) as defined by organizational policy. Also, adopting a process for multiple surgical counts of instruments, sharps, and sponges intraoperatively is crucial to prevent the accidental retention of foreign objects, which can lead to life-threatening complications. Lastly, monitoring surgical and procedural errors provides an opportunity to implement data-driven corrective measures accordingly. Such monitoring could be through a set of performance indicators, audit rounds, chart reviews, or staff feedback reports.*	
Evidence:	
9.3.1	*A written policy outlines surgical and procedural safety aligned with evidence-based guidelines and best practices.*
9.3.2	*Evidence of adopting and implementing a standardized surgical safety checklist that covers the sign-in, time-out, and sign-out verification phases.*
9.3.3	*Documented records of using multiple surgical counts of instruments, sharps, and sponges intraoperatively.*
9.3.4	*Evidence of reporting and analyzing surgical safety events and developing corrective action plans to address these events.*
9.3.5	*Evidence of reviewing and monitoring surgical and procedural errors, with an action plan developed accordingly.*

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Criteria 9.4: the organization implements evidence-based practices to enhance diagnostic accuracy.

Description:

This criterion underlines the methodical use of evidence-based practices to enhance diagnostic accuracy and reduce the likelihood of misdiagnoses. This includes adopting interventions such as proper patient evaluation, advanced diagnostic technologies, standardized interpretation criteria, and regular proficiency assessments. For instance, embracing technologies, such as clinical decision support systems, enhances diagnostic accuracy through up-to-date diagnostic solutions. Further, adopting or adapting checklists (e.g. radiology reporting checklist), algorithms (e.g. stroke evaluation algorithms), protocols (e.g. chest pain protocol), or clinical pathways (e.g. sepsis pathway) provide a standardized framework to ensure diagnostic accuracy. Moreover, integrating case discussions and clinical reasoning exercises into medical staff meetings promotes knowledge sharing and a collaborative problem-solving approach. In this, clinical reasoning exercises are structured activities that enhance healthcare providers' clinical reasoning skills by analyzing patient data, considering potential diagnosis options, and making informed clinical decisions. Similarly, establishing a structured peer review process to verify diagnoses is important to detect inaccuracies and enhance diagnostic precision. Lastly, monitoring diagnostic errors, the timeliness of diagnoses, and the consultation promptness assist in error identification and correction. Such monitoring could be through a set of performance indicators, audit rounds, chart reviews, or staff feedback reports.

Evidence:

9.4.1	*Evidence of embracing advanced diagnostic technologies such as clinical decision support systems or artificial intelligence-assisted diagnostic tools.*
9.4.2	*Evidence of adopting or adapting checklists, algorithms, protocols, or clinical pathways to guide evidence-based diagnostic practices.*
9.4.3	*Evidence of using case discussions and clinical reasoning exercises in medical staff meetings.*
9.4.4	*Evidence of implementing a structured peer review process to impartially and periodically review the accuracy of diagnostic decisions.*
9.4.5	*Evidence of reviewing and monitoring diagnostic errors, timely diagnosis, and timely request and response of second opinions and consultations.*

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Criteria 9.5: the organization participates in conducting patient safety research to contribute to overall patient safety.	
Description: *This criterion fosters the commitment to advance patient safety through participation in research. Organizational participation in patient safety research could be through organizing, collaborating, funding research projects, and sharing patient safety data with researchers or organizations while maintaining confidentiality. The pursuit of patient safety research seeks to play a role in developing practices that can be shared widely to contribute to overall patient safety. In this, healthcare organizations need to establish a structured policy that guides the process for initiating, approving, and conducting patient safety-related research projects. Such a policy is crucial to ensure that research projects are conducted rigorously and ethically. Equally important is to equip the staff involved in patient safety research with an understanding of research processes and ethical principles. Evidence of recent or ongoing patient safety research and evidence of sharing the research results with the broader healthcare community through publications, posters, or presentations enhance patient safety practices. Lastly, monitoring patient safety research projects to ensure adherence to ethical principles and progress timelines is essential. Such monitoring could be through discussion in the dedicated committee(s), audit rounds, research progress reports, research protocol amendment reports, staff feedback reports, or customer feedback reports.*	
Evidence:	
9.5.1	*A written policy or guide outlines the process of initiating, approving, and conducting patient safety-related research projects.*
9.5.2	*Records of educating staff members involved in patient safety research on the research process and ethical principles.*
9.5.3	*Evidence of ongoing or completed patient safety research within the last 18 months, including the project description, research method, and timeline.*
9.5.4	*Evidence indicates sharing patient safety research publications, posters, or presentations with the broader healthcare community.*
9.5.5	*Evidence of reviewing and monitoring the adherence of patient safety research to ethical principles and progress timeline.*

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Domain 10: Patient Safety Preventive Measures

Preventing patient safety events is a fundamental safety commitment in healthcare. It is a continuous effort, focusing on various aspects of patient care to ensure safety, quality, and equity [104]. Hence, healthcare organizations aim to minimize misidentification errors, prevent Healthcare-Associated Infections (HAIs), reduce pressure injuries, avoid patient falls, and address healthcare disparities while promoting equitable patient safety [105]. For instance, healthcare organizations employ evidence-based practices to prevent misidentification by accurately identifying patients to ensure that patients receive the right care at the right time and by the right healthcare providers, using techniques such as barcode scanning and unique identifiers [106].

Also, healthcare organizations adopt evidence-based strategies and guidelines to prevent and control HAIs. This includes rigorous hand hygiene practices, sterilization protocols, and the proper use of personal protective equipment [107]. To ensure understanding and adherence to these practices, regular staff training and continuous monitoring of infection rates are crucial components [108]. Preventing HAIs is not only about protecting individual patients but also contributes to broader public health goals by reducing the spread of infections.

Further, healthcare organizations dictate various evidence-based practices to assess and prevent pressure injuries and patient falls. These practices started initially by assessing the risk of developing these safety events. Thereafter, healthcare organizations deploy multiple practices, such as regular repositioning, appropriate nutrition, and specialized equipment when necessary to prevent pressure injuries [109]. Also, it is recommended that bed alarms be installed or mobility aids be provided to reduce patient falls [110]. These measures, backed by ongoing staff training and monitoring, enhance patient outcomes and reduce unnecessary healthcare costs associated with the complications of pressure injuries and patient falls.

Last but not least, preventing healthcare disparities and promoting equitable patient safety are vital components of the mission of healthcare organizations [111]. Therefore, organizations collect and analyze patient demographic data to identify disparities in healthcare outcomes. This analysis is followed by developing and implementing various evidence-based practices and interventions to address identified healthcare disparities. These practices include providing culturally sensitive care, offering language services, and adopting community outreach programs [112]. Health organizations can promote equitable patient safety by engaging with patient and community advocacy groups to ensure that care is accessible and patient-centered [113].

In conclusion, preventing patient safety events in healthcare organizations is a comprehensive and interconnected effort that includes several care areas. By diligently applying evidence-based practices, organizations can create a safer and more equitable healthcare environment for patients. This subsequently fulfills the organizational commitment to providing high-quality and safe healthcare for every patient they serve.

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Criteria 10.1: the organization implements evidence-based practices to prevent errors due to misidentification.	
Description: *This criterion underscores the importance of implementing evidence-based practices in the patient identification process to prevent errors associated with misidentification. To begin, healthcare organizations must establish a policy or protocol that outlines the patient identification process, method, and procedure to standardize this critical aspect of healthcare delivery. Simultaneously, orienting staff on deploying this policy is important to ensure consistent implementation of correct patient identification procedures. In addition, implementing robust strategies in patient identification processes, such as identification tools, using technologies, and establishing clear communication protocols, will enhance patient safety and reduce misidentification errors. For instance, adopting evidence-based technology, such as barcode scanning, for patient identification is a step that minimizes human errors in patient identification. Also, a standardized practice of using at least two identifiers during patient interactions adds an extra protection layer to the identification process. Yet, the journey doesn't end with implementation; it extends to the ongoing review and monitoring of compliance with the patient identification policy and process. This practice ensures that the organization remains responsive to emerging best practices in the identification process. Such monitoring could be through a set of performance indicators, regular observational audits, staff feedback reports, or customer feedback reports.*	
Evidence:	
10.1.1	*A written policy or protocol outlines the patient identification process, procedure, and responsibility.*
10.1.2	*Evidence of integrating patient identification methods in the orientation program of newly hired staff.*
10.1.3	*Evidence of using evidence-based technology for patient identification, such as barcode scanning.*
10.1.4	*Evidence of a standardized use of at least two identifiers during patient interactions.*
10.1.5	*Evidence of reviewing and monitoring compliance with patient identification policy and process at least quarterly.*

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Criteria 10.2: the organization implements evidence-based practices to prevent Healthcare-Associated Infections (HAIs).	
Description: *This criterion underscores the importance of using evidence-based practices to identify, prevent, mitigate, and manage Healthcare-Associated Infections (HAIs), thereby reducing the risk of infection transmission. Healthcare organizations need to prioritize infection prevention and control by establishing a program or policy that outlines infection prevention, surveillance, and control strategies. This policy offers a structured and evidence-based framework for all healthcare providers. Furthermore, staff training and competency assessments related to infection prevention strategies cannot be overstated. It ensures that all personnel have the knowledge and skills to implement infection control strategies to prevent HAIs effectively. A bundle of interventions may assist in reducing HAIs, such as hand hygiene, aseptic technique, infection control precautions, surveillance systems, antibiotic stewardship, and environmental hygiene. Hand hygiene is a critical frontline defense against the spread of infections. Also, implementing antimicrobial stewardship strategies combat the rise of antimicrobial resistance, which subsequently mitigates HAI risks. Lastly, monitoring the occurrence of HAIs allows for identifying trends and areas for improvement in HAIs. Such monitoring could be through a set of performance indicators, inspection reports, surveillance reports, dashboards, or staff feedback reports.*	
Evidence:	
10.2.1	*A written policy or program outlines the infection prevention, surveillance, and control strategies.*
10.2.2	*Records of staff training or competency assessments related to infection prevention strategies.*
10.2.3	*Reports indicate a 75% or more overall compliance with hand hygiene practices.*
10.2.4	*Evidence of implementing and documenting antimicrobial stewardship strategies.*
10.2.5	*Evidence of reviewing and monitoring surveillance reports on the occurrence of healthcare-associated infections at least quarterly.*

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Criteria 10.3: the organization implements evidence-based practices to prevent pressure injuries.	
Description: *This criterion underlines the importance of implementing evidence-based practices to prevent pressure injuries, which can cause patient discomfort and complications. To this end, healthcare organizations need to establish a policy that outlines pressure injury risk assessment and prevention strategies. Additionally, conducting training or competency for healthcare staff on pressure injury risk assessment and prevention strategies ensures effective implementation. In parallel, incorporating various strategies such as using prevention protocols, using risk assessment tools, adequate patient repositioning, and providing specialized support surfaces enables healthcare providers to identify at-risk patients, implement preventive measures, and manage pressure injuries, ultimately enhancing safety while reducing the costs associated with a pressure injury. For instance, conducting pressure injury risk assessments using validated tools is a key step in identifying at-risk patients and customizing preventative measures. Also, implementing evidence-based support surfaces designed specifically to prevent pressure injuries is a proactive measure that minimizes the likelihood of pressure injuries occurring. Yet, pressure injury prevention doesn't end with implementation; it extends to monitoring pressure injury rates and compliance with prevention strategies. Such monitoring could be through a set of performance indicators, inspection reports, audit rounds, chart reviews, staff feedback reports, or customer feedback reports.*	
Evidence:	
10.3.1	*A written policy or guideline outlines the pressure injury risk assessment process, procedure, and prevention strategies.*
10.3.2	*Records of nursing staff training or competency assessments on pressure injury prevention strategies.*
10.3.3	*Evidence of conducting and documenting pressure injury risk assessments using validated tools.*
10.3.4	*Evidence of using evidence-based support surfaces designed to prevent pressure injuries.*
10.3.5	*Evidence of reviewing and monitoring pressure injury rate and compliance with prevention strategies at least quarterly.*

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Criteria 10.4: the organization implements evidence-based practices to prevent patient falls.	
Description: *This criterion emphasizes the importance of implementing evidence-based practices to prevent patient falls, a common and potentially serious risk in healthcare settings. To this end, healthcare organizations need to establish a policy that outlines patient fall risk assessment and prevention strategies. Additionally, conducting training or competency for healthcare staff on patient fall risk assessment and prevention strategies ensures the effective implementation of these strategies. In parallel, incorporating various strategies such as prevention protocols, fall risk assessment tools, optimal lighting, clutter-free environments, and assistive devices enables healthcare providers to identify at-risk patients, implement preventive measures, and respond effectively to potential fall incidents. For instance, implementing patient fall risk assessments using validated tools is key in identifying those at risk and tailoring interventions accordingly. Also, using evidence-based assistive devices for at-risk patients is a proactive measure that can significantly reduce the likelihood of falls. Yet, it's not enough to implement these measures once; ongoing monitoring of patient fall rates and compliance with fall prevention strategies are essential to identify trends and areas for improvement. Such monitoring could be through a set of performance indicators, inspection reports, audit rounds, chart reviews, staff feedback reports, or customer feedback reports.*	
Evidence:	
10.4.1	*A written policy or guideline outlines the patient fall risk assessment process, procedure, and prevention strategies.*
10.4.2	*Records of nursing staff training or competency assessments on patient fall prevention strategies.*
10.4.3	*Evidence of conducting and documenting patient fall risk assessments using validated tools.*
10.4.4	*Evidence of using evidence-based assistive devices for patients at risk of falling.*
10.4.5	*Evidence of reviewing and monitoring patient fall rate and compliance with fall prevention strategies at least quarterly.*

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Criteria 10.5: the organization implements evidence-based practices to prevent healthcare disparities and promote equitable patient safety.	
Description: *This criterion highlights the importance of addressing healthcare disparities to enhance equitable and tailored patient care. In this, healthcare organizations need to establish a policy or guideline that outlines the systematic collection, analysis, and prevention of healthcare disparities. Explicitly, organizations need to identify disparities by collecting and analyzing data on patient outcomes, followed by stratifying data by socio-demographic characteristics to identify any disparities along with the root causes of these disparities. Thereafter, design evidence-based interventions to address and reduce inequalities and foster an equity culture, such as using multilingual education materials and interpreter services. To implement the policy, training healthcare providers and leaders on health disparities and cultural competency is vital in ensuring that the entire healthcare team is equipped to tackle these issues sensitively and effectively. The training may include bias awareness, cultural sensitivity, and social determinants of health. But the journey doesn't end there. The ongoing monitoring of the action plan implementation is essential to ensure that the organization remains responsive and adaptive to the evolving disparities and adjusts as necessary. Such monitoring could be through a follow-up action plan tracking records, observational reports, staff feedback reports, or customer feedback reports.*	
Evidence:	
10.5.1	*A written policy or guide outlines the process, procedure, timeframe, and responsibility of collecting, analyzing, and preventing health disparities.*
10.5.2	*Records of staff and leaders training on healthcare disparities and cultural competency.*
10.5.3	*A report indicates assessing and identifying the causes of disparities, followed by an action plan.*
10.5.4	*Evidence of implementing evidence-based interventions to address and reduce disparities.*
10.5.5	*Evidence of reviewing and monitoring the action plan implementation and adjusting the plan as needed.*

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